

PG SOFT

Integration Document

Seamless Mode v2.4.11

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Revision History

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		• 3.1.5 Others • 9.2 Currencies Amended headings & description • 3.1.1.2 GetLaunchURLHTML API • 3.1.1.3 Reality Check • 3.2 Sample Request • 3.3 Game Launch Restrictions • 5.1.3.1 Get Player Wallet • 5.1.3.2.1 Bet (Deprecated) • 5.1.3.2.2 Payout (Deprecated) • 5.1.3.2.3 Bet Payout • 5.1.3.3 Balance Adjustment Remove description • 5.1.1.3 Hash Authentication
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		Remove currency JPY & GBP 9.3 Currencies Remove parameter iwk 3.1.1.2 GetLaunchURLHTML API Remove Contents - Bet (Deprecated) - Payout (Deprecated) Add new content 3.1.1.3 Troubleshooting 3.2 Sample Request 3.4 FAQ Remove Hong Kong & United Kingdom 8.1 IP
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1 Overview

This document details the Integration of the operator's wallet mode, also known as Seamless Wallet mode. This process details that PG SOFT system will call the operator API in real time for every single transaction while the player is playing games to achieve the addition or deduction of the player's balance.

2 System Requirements

2.1 Game Dimensions

PG SOFT games are responsive H5 and are displayed vertically, thus, minimum dimensions are required. Please refer to the following for game dimensions in iframe:

- Ideal dimensions : 1280 x 720
- Minimum dimensions : 540 x 960
- Maximum dimensions : 1080 x 2340

2.2 Hardware

	Android	iOS	Windows
Version of Operating System	5.0 or above	11.0 or above	7 or above
RAM	2GB	2GB	2GB
CPU	ARM Cortex-A9 or above	iPhone 7 or above	• 1 gigahertz (GHz) or faster • 32-bit (x86) or 64-bit (x64) processor • Dual core at least
Storage	2GB and above	2GB and above	20GB and above

2.3 Browser

- Microsoft Edge : version 79 or above
- Firefox : version 58 or above
- Google Chrome : version 59 or above
- Safari : version 11.1 or above
- *others* : browser based on Chromium that equal to Google Chrome 59 or above

2.4 Unsupported Browser

- Android : Firefox browser, CM browser
- iOS : CM browser
- Desktop : UC browser, Baidu browser, Quark browser, CM browser, IE browser

3 Launch Game

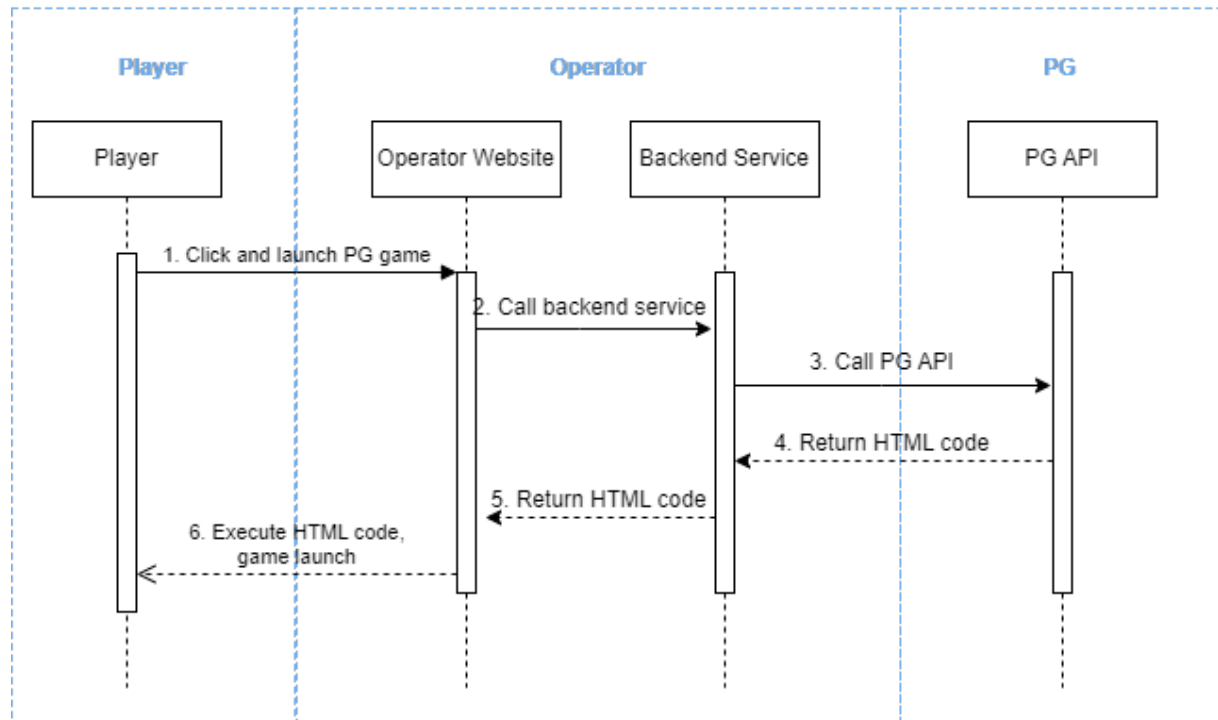
3.1 Game Authorization Modes

PG SOFT supports one kind of game launch authorization mode:

- **HTML scheme** mode is for iOS, Android and browser, it uses the token parameters to authorize and login.

3.1.1 HTML Scheme

HTML scheme is the conventional method for launching the game. Players need to log in through the operator's website and start the game via the game lobby.



1. Players will visit the operator's website and click on the PG SOFT game.
2. The operator's frontend will call the backend service to obtain the HTML code.
3. In response to the frontend's request, the backend service will call the PG SOFT GetLaunchURLHTML API.
4. The PG SOFT API returns the HTML code.
5. The backend will send this HTML code to the frontend.
6. Upon receiving the HTML code, the frontend will execute it, displaying the HTML content on the player's browser (running PG SOFT game).

*** Note:** Please pass the HTTP Header and Body response from Step 4 of PG SOFT directly to the player's browser in Step 5, without any additional processing, to prevent unexpected errors.

The Advantage:

PG SOFT will return one or more game launch URL based on the request parameters (e.g., client_ip, group_id, etc.). The best game launch URL will be selected for play by detecting on the player's side.

In the event of domain name issues, PG SOFT can unilaterally switch the game server address without requiring cooperation from the operator, thus reducing the technical workload on their teams.

Why deprecate URL Scheme?

1. In today's complex network landscape, there is currently no service provider in the market capable of perfectly simulating the diverse network conditions experienced by each player. Traditional solutions have proven ineffective in addressing the various network issues encountered by players.
2. To guarantee the optimal gaming experience for every player, PG SOFT is introducing an HTML-based solution. We will test every game launch URL returned from the API, examine it from the player's perspective, ensuring that our testing process remains transparent to operators while alleviating the technical workload on their teams.
3. For the diverse network conditions that may arise at any time across different regions (e.g., domain blocking and domain hijacking, etc.), PG SOFT will continuously fine-tune and optimize our HTML code based on real-world scenarios.
4. To enhance domain security and ensure uninterrupted operations for operators, PG SOFT has also implemented HTML encryption to prevent potential domain blocking and domain hijacking by third parties.

3.1.1.1 Implementation

STEP 1

The operator needs to call the PG SOFT GetLaunchURLHTML API through its own backend to obtain the HTML page code.

*** Note:** Operator should include the following (headers) in the API response to prevent the HTML response from being stored in the web browser cache:

Cache-Control: no-cache, no-store, must-revalidate

Header Parameters:

*** Note:** The following are for reference only and might update accordingly.

HTTP Header	Remarks
Content-Type: text/html	HTTP Body format of the response
Content-Length	HTTP Body length of the response
Transfer-Encoding: chunked	The server may also use Transfer-Encoding to indicate the length of the returned content. Majority of clients still support this format.
Content-Encoding: gzip	Gzip compression is used in the HTTP Body of the response
Cache-Control: no-cache, no-store, must-revalidate	Browser cache control to prevent the browser from caching the HTTP Body of the response

Header Examples:

```
200 OK HTTP/1.1
Content-Type: text/html
Content-Encoding: gzip
Content-Length: 2910
Cache-Control: no-cache, no-store, must-revalidate
Server: nginx/1.24.0
Date: Thu, 16 Nov 2023 07:43:39 GMT
```

STEP 2

After obtaining the HTML page code, execute the HTML code through a web browser.

Please refer to the following operator's lobby as an example:

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.6.4/jquery.min.js">
</script>
<script>
    $(document).ready(function(){
        $("button").click(function(){
            $.post( "https://<Operator backend service>/api",
                { param1: "value1", param2: "value2" },
                function(data) {
                    var newPage = window.open('', '_blank');
                    newPage.document.open();
                    newPage.document.write(data);//Execute HTML Code
                    newPage.document.close();
                });
        });
    });
</script>
</head>
    <body><button>Launch Game</button></body>
</html>
```

3.1.1.2 GetLaunchURLHTML API

The operator needs to call the PG SOFT GetLaunchURLHTML API through its own backend to obtain the HTML page code.

Request

API URL : {PgSoftAPIDomain}/external-game-launcher/api/v1/GetLaunchURLHTML

HTTP method : POST

URL Parameters:

Parameter name	Data Type	Mandatory	Description
trace_id	String	Yes	Unique identity (GUID) of the request Note: Please set the parameter value as GUID format

Body Parameters:

Parameter name	Data Type	Mandatory	Description
operator_token	varchar(20)	Yes	Unique identity of operator
path	varchar(500)	Yes	Game path: /<GameID>/index.html Web Lobby path: /web-lobby/{panel_type}/ Tournament path: /tournament/ Bet History Interpreter path: /history/redirect.html Note: - Please encode the value with UrlEncode to avoid unexpected errors - GameID is the Unique identity for each game¹ - panel_type Display panel of web lobby at first load [NO VALUE] : Default Panel (Tournament) tournament/open : Tournament – Open for entry panel tournament/process: Tournament – In progress panel

¹ Please get the latest list from PG SOFT

			tournament/end : Tournament – Ended panel games : Show only game
extra_args	Query String Parameter	Yes	Launch game and bet history interpreter parameter (Please refer to relevant parameter list) Note: • Please encode the value with UrlEncode to avoid unexpected errors • Maximum 500 characters
url_type	varchar(50)	Yes	URL Type: game-entry: Game entry web-lobby : Game web lobby entry web-tournament: Tournament entry web-history: Bet history interpreter
client_ip	varchar(50)	Yes	Player IP
group_id	UInt16	No	Operator group ID

extra_args Parameters (Game Launch) :

Parameter name	Data Type	Mandatory	Description
btt	UInt8	Yes	Game launch mode ²
ops	varchar(200)	Yes	Token generated by operator system Note: • Please encode the value with UrlEncode to avoid unexpected errors • Max 200 characters • Please set the parameter value as GUID format or any random string
l	varchar(50)	No	Display language of game ³ Default: en
te	UInt16	No	Time elapsed for game health reminder in seconds for operator (Reality check ⁴)

² Please refer to [Bet Types](#) section for complete list

³ Please refer to [Languages](#) section for complete list

⁴ Please refer to [Reality Check](#) section for more information

ri	UInt16	No	Reminder interval for game health reminder in seconds (Reality check ⁴)
op	String	No	Custom parameter for operator, PG SOFT API will include parameter value upon verifying operator's player session Note: Please encode the value with UriEncode to avoid unexpected errors
f	String	No	Game exit URL Default: Redirects to PG SOFT quit page Note: • Please encode the value with UriEncode to avoid unexpected errors • Assign value to <i>PGGameCloseUrl</i> to close the game window
oc	Integer	No	Screen orientation check ⁵ when launching the game 0: Skip screen orientation check 1: Normal mode (Default)

extra_args Parameters (Web Lobby):

Parameter name	Data Type	Mandatory	Description
ops	varchar(200)	Yes	Token generated by operator system Note: • Please encode the value with UriEncode to avoid unexpected errors • Max 200 characters • Please set the parameter value as GUID format or any random string
l	varchar(50)	No	Display language ⁶ of web lobby Default: en
width	Integer	No	Page width of web lobby Default: Responsive Value range: Value ≥ 930

⁵ Please refer to screenshot

⁶ Please refer to [Languages](#) section for complete list

extra_args Parameters (Tournament):

Parameter name	Data Type	Mandatory	Description
ops	varchar(200)	Yes	Token generated by operator system Note: • Please encode the value with UrlEncode to avoid unexpected errors • Max 200 characters • Please set the parameter value as GUID format or any random string
l	varchar(50)	No	Display language of game ⁶ Default: en

extra_args Parameters (Bet history interpreter⁷) :

Parameter name	Data Type	Mandatory	Description
t	String	Yes	Operator session
psid	String	Yes	Parent Bet ID
sid	String	Yes	Bet ID
gid	String	No	Game ID
lang	String	No	Language: • en (default) • zh
type	String	Yes	Fixed value: operator

Example:

Body:

operator_token=xxxx&path=%2F1%2Findex.html&extra_args=btt%3D1%26ops%3D12345-abcd-1234-abcd-12345&url_type=game-entry&client_ip=1.1.1.1

⁷ Please refer to [Bet History Interpreter](#) section for complete information

Response

HTTP Body response:

The browser only needs to execute the HTML page to launch the game.

*** Note: The HTML code is subject to adjustment and optimization, and does not follow a fixed format. Please avoid using deserialization or parsing methods to retrieve the final game launch URL.**

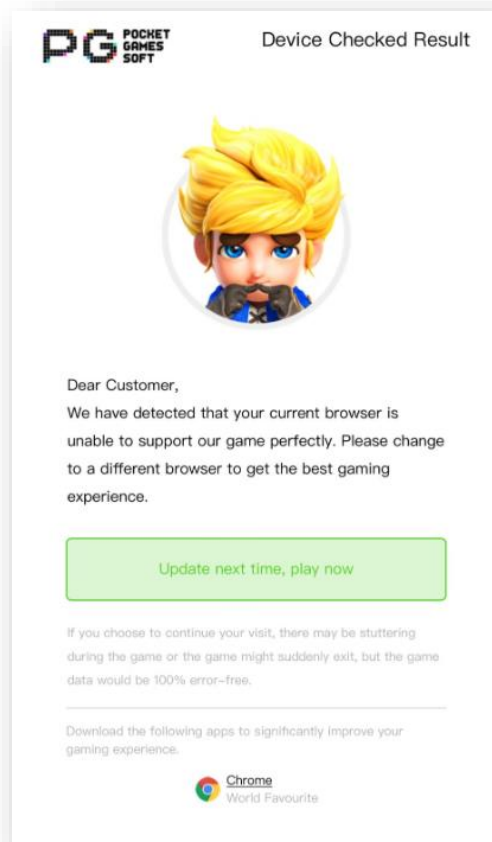
HTTP status code response:

Status Code	Description	Reason	Solution
200	Successful response	-	-
400	Invalid Parameters	The format of the parameter is not provided as requested	Check whether the HTTP Method, HTTP Header, and HTTP Body are requested accordingly to the requirements.
404	Path not found	Invalid path	Check whether the path is correct.
500	Server Internal Error	An internal server error occurred	• Retry • Inform PG SOFT
504	Gateway Timeout	The request has failed to reach the destination server successfully and a proxy server failed.	• Retry • Inform PG SOFT

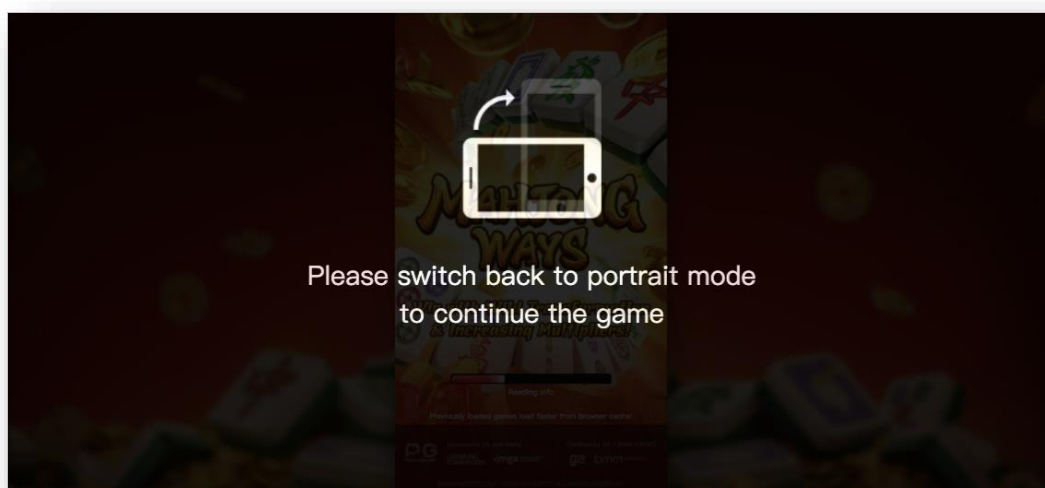
Example for HTML Response (For reference only):

```
<!doctype html><html lang="en">...</html>
```

Screenshot



Device compatibility check



Screen orientation

3.1.1.3 Troubleshooting

The following scenarios may prevent players from successfully accessing the game:

- Domain name resolution failure (e.g., due to hijacking, blocking, or other interferences)
- PG SOFT network issues
- Player-side network failure
- Outdated system version (may cause SSL certificate verification to fail)
- Browser restrictions on JavaScript execution
- Browser settings preventing image downloads
- Browser restrictions on redirection
- Ad-blocking plugins installed in the browser
- Use of a non-mainstream or outdated browser

3.1.1.4 Reality Check

In HTML scheme⁸, operator may include “te” and “ri” parameter into the launch game URL for reality check.

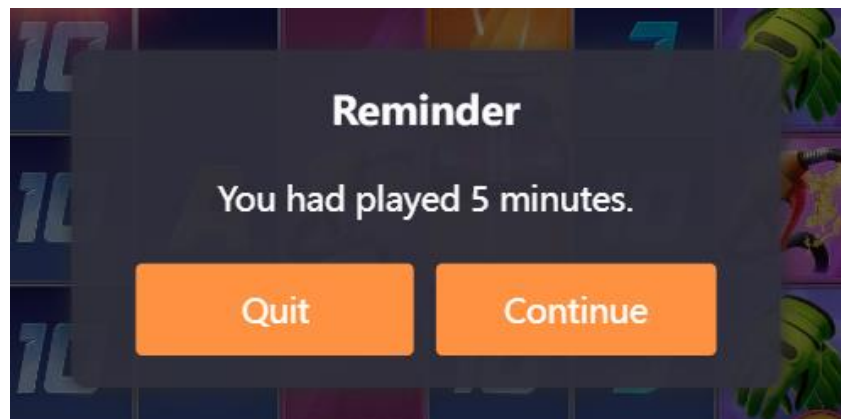
- te : Time used in the calculation for the first reality check in second(s)
- ri : Reality check interval in second(s)

Example:

ri=300&te=290

The first reality check reminder will show after 10 seconds (300s-290s=10s), and subsequent reminders will show up for every 5 minutes.

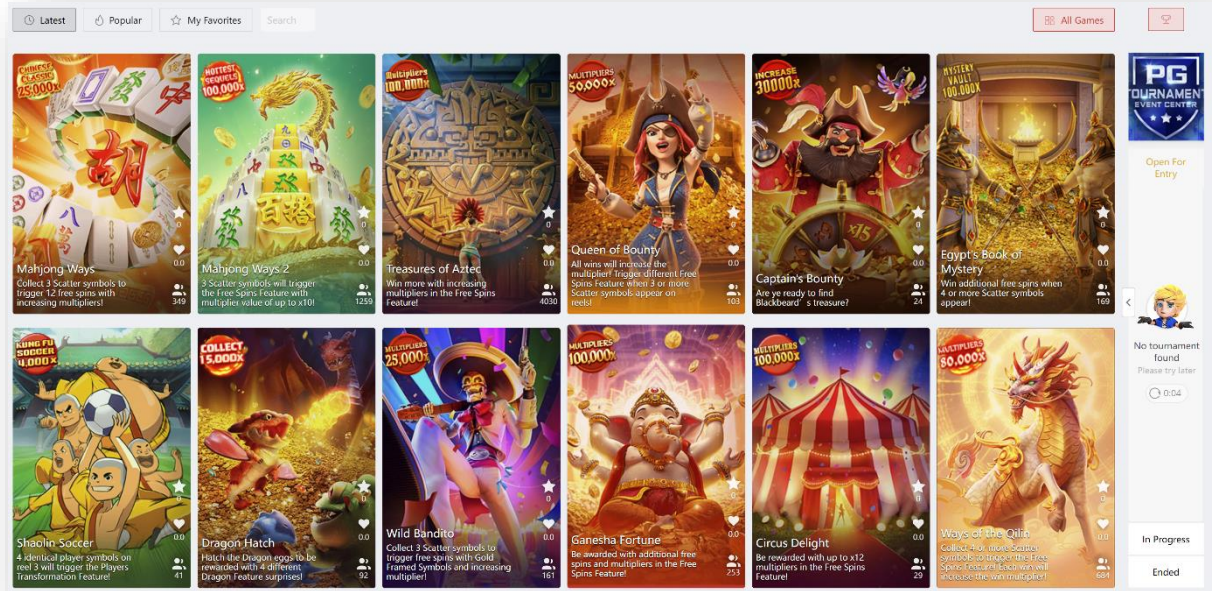
Screenshot



Health reminder

⁸ Please refer to [HTML Scheme](#) section for more information

Screenshot



Web Lobby

3.1.2 Android SDK Scheme

3.1.2.1 Implementation

STEP 1: Preconfigured

1. Get the SDK AAR file from PG SOFT.
2. Call the API⁹ to get a JSON response.

Note:

Staging SDK file: <https://app.pg-staging.com/game-sdk-android/game-sdk.zip>

* Note: Production SDK file please request from PG SOFT support.

* Note: Operator should include the following (headers) in the API response to prevent the HTML response from being stored in the web browser cache:

Cache-Control: no-cache, no-store, must-revalidate

Header Parameters:

* Note: The following are for reference only and might update accordingly.

HTTP Header	Remarks
Content-Type: text/json	HTTP Body format of the response
Content-Length	HTTP Body length of the response
Transfer-Encoding: chunked	The server may also use Transfer-Encoding to indicate the length of the returned content. Majority of clients still support this format.
Content-Encoding: gzip	Gzip compression is used in the HTTP Body of the response
Cache-Control: no-cache, no-store, must-revalidate	Browser cache control to prevent the browser from caching the HTTP Body of the response

Header Examples:

```
200 OK HTTP/1.1
Content-Type: application/json
Content-Length: 2910
Cache-Control: no-cache, no-store, must-revalidate
Server: nginx/1.24.0
Date: Thu, 16 Nov 2023 07:43:39 GMT
```

⁹ Please refer to [GetLaunchURLSDK API](#) section for more information

STEP 2: Import SDK and dependencies

1. Import [dnsjava](#) dependency.
2. Import the SDK Android AAR file into the project:

```
// build.gradle
dependencies {
    implementation files("libs/pg-gamesdk-release.aar")
    implementation group: 'dnsjava', name: 'dnsjava', version: '3.5.2'
    ...
}
```

3. (Recommended) Android builder.gradle minSdk is set to at least **26**.

STEP 3: Apply the Android SDK

Launch game through GameWebViewFragmentActivity

The SDK provides GameWebViewFragmentActivity to allow users to launch games quickly.

1. Call the API¹⁰ to get a JSON response and build an SDK webview configuration through GameWebViewConfigModel.Builder.
2. Use GameWebViewConfigModel.Builder to build a DNS configuration for the SDK.
3. Pass the DNS and GameWebView configuration to GameWebViewFragmentActivity through intent.putExtra(<GameWebViewFragmentActivity.KEY>,Bundle) to launch the game:

```
// Main Activity

// data return from API
String response = "{\"error\":null,\"data\":...}"
Intent intent = new Intent(MainActivity.this,
GameWebViewFragmentActivity.class);

Bundle gameWebViewConfigBundle = new GameWebViewConfigModel.Builder()
                                .sdkResponse(responseData)
                                .locale(GameSDKLocale.ZH)
                                .toBundle();
intent.putExtra(GameWebViewFragmentActivity.GAME_WEB_VIEW_CONFIG_BUNDLE,
gameWebViewConfigBundle);

Bundle dnsConfigBundle = DNSConfigInstance.GOOGLE_DNS_SERVER.getBundle();
intent.putExtra(GameWebViewFragmentActivity.DNS_CONFIG_BUNDLE,
dnsConfigBundle);

getApplicationContext().startActivity(intent);
```

Custom GameWebViewFragmentActivity

To load GameWebView through FragmentManager in FragmentActivity, please refer the following code:

*** Note: GameWebViewFragment needs to be committed in the callback after the onCreate life cycle.**

¹⁰ Please refer to [GetLaunchURLSDK API](#) section for more information

```
// CustomFragmentActivity / CustomAppCompatActivity
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.demo_fragment_activity);
}

@Override
protected void onStart() {
    GameWebViewFragment webViewFragment = new
    GameWebViewFragment(DNSConfigInstance.GOOGLE_DNS_SERVER, GameSDKLocale.ZH);

    getSupportFragmentManager().beginTransaction()
        .add(R.id.demo_fragment_container, webViewFragment,
    GameWebViewFragment.class.getName())
        .commitNow();

    // data return from API
    String response = "{\"error\":null,\"data\":...}"
    webViewFragment.loadGame(response);
}

// remove GameWebViewFramgment and clean up
private void removeView() {

    getSupportFragmentManager().beginTransaction().remove(webViewFragment).comm
    itNow();
    webViewFragment = null;
}
```

```
// demo_fragment_activity.xml
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/demo_fragment_container"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
</androidx.constraintlayout.widget.ConstraintLayout>
```

Custom DNS

The SDK provides four DNS servers by default to resolve domains:

```
DNSConfigInstance.GOOGLE_DNS_SERVER;  
DNSConfigInstance.CLOUDFLARE_DNS_SERVER;  
DNSConfigInstance.TENCENT_DNS_SERVER;  
DNSConfigInstance.ALI_DNS_SERVER;
```

If you want to use other custom DNS servers, you can use `DNSConfigModel.Builder` to create the DNS configuration required by the SDK.

If you wish to understand more about DNS resolution and DoH (DNS over HTTPS), please refer the following link:

- [What is DNS](#)
- [CloudFlare public DNS address](#)
- [Apply CloudFlare DNS over HTTPS](#)

The IP and DoH domain must be different values.

Only PG SOFT internal products' domains will be resolved into IP by DoH, other domains such as google.com will be resolved by public DNS.

```
DNSConfigModel customDnsModel = new DNSConfigModel.Builder()  
    .ip("1.1.1.1")  
    .domain("https://1.0.0.1/dns-query")  
    .headerMap(new HashMap<String, String>()  
    {{  
        put("Accept", "application/dns-json");  
    }}).build();  
GameWebViewFragment webViewFragment = new GameWebViewFragment(customDnsModel);  
...
```

SDK exception callback

- GameWebViewFragment provides setOnErrorReceive to get GameWebView exceptions, VPN configuration exceptions or GameSDKResponseModel exceptions.
- GameSDKJsonResponseException is thrown by GameSDKResponseModel. This exception is usually thrown when the response data content is incorrect.
- GameSDKSetupError is thrown by GameWebView or GameVPNService. This exception is usually thrown when DNSConfigModel or GameWebViewConfigModel pass invalid or wrong values.
- GameSDKCommonError is thrown by GameWebViewFragment or GameVPNService. This exception is usually thrown when data verification passed with incorrect data, such as invalid DNS Server, Game URL.

```
webViewFragment.setOnErrorReceive(new GameWebViewErrorNotify() {  
    @Override  
    public void onErrorReceive(Exception e) {  
        if (e instanceof GameSDKJsonResponseException) {  
            e.printStackTrace();  
        } else {  
            throw new RuntimeException(e);  
        }  
    }  
});
```

SDK localization

- Since there is no unified standard for application localization and Locale settings before [Android 13](#), GameSDK provides GameSDKLocale to allow developers to set the language used in GameWebView.
- The SDK provides language codes (en, zh) in accordance with ISO 639-1 and is packaged using GameSDKLocale.
- GameSDKLocale language code table can be found [here](#).
- Please refer to the Locale static variable provided in GameSDKLocale for the list of languages currently provided by the SDK.

```
// if language code not support, it will return GameSDKLocale.EN by default.  
GameSDKLocale locale = GameSDKLocale.getGameSDKLocale("en");  
new GameWebViewConfigModel.Builder()  
    .sdkResponse(responseData)  
    .locale(locale)  
    .build();  
  
// or direct set while new GameWebViewFragment  
GameWebViewFragment webViewFragment = new  
GameWebViewFragment(DNSConfigInstance.GOOGLE_DNS_SERVER, GameSDKLocale.EN);
```

3.1.2.2 GetLaunchURLSDK API

The operator needs to call the PG SOFT GetLaunchURLSDK API through its own backend to obtain the encrypted redirection address data in JSON format.

Request

API URL : {PgSoftAPIDomain}/external-game-launcher/api/v2/GetLaunchURLSDK

HTTP Method : POST

URL Parameters:

Parameter Name	Data Type	Mandatory	Description
trace_id	String	Yes	Unique identity (GUID) of the request Note: Please set the parameter value as GUID format

Body Parameters:

Parameter Name	Data Type	Mandatory	Description
operator_token	varchar(20)	Yes	Unique identity of operator
path	varchar(500)	Yes	Game path: /<GameID>/index.html Web Lobby path: /web-lobby/{panel_type}/ Tournament path: /tournament/ Bet History Interpreter path: /history/redirect.html Note: - Please encode the value with UrlEncode to avoid unexpected errors GameID is the Unique identity for each game¹¹ panel_type Display panel of web lobby at first load [NO VALUE] : Default Panel (Tournament) tournament/open : Tournament – Open for entry panel

¹¹ Please get the latest list from PG SOFT

			tournament/process: Tournament – In progress panel tournament/end : Tournament – Ended panel games : Show only game
extra_args	Query String Parameter	Yes	Launch game parameter (Please refer to relevant parameter list) Note: ● Please encode the value with UrlEncode to avoid unexpected errors ● Maximum 500 characters
url_type	varchar(50)	Yes	URL Type: game-entry: Game entry web-lobby: Game web lobby entry web-tournament: Tournament entry
client_ip	varchar(50)	Yes	Player IP
group_id	UInt16	No	Operator group ID

extra_args Parameters (Game Launch):

Parameter Name	Data Type	Mandatory	Description
btt	UInt8	Yes	Game launch mode ¹²
ops	varchar(200)	Yes	Token generated by operator system Note: ● Please encode the value with UrlEncode to avoid unexpected errors ● Max 200 characters ● Please set the parameter value as GUID format or any random string
l	varchar(50)	No	Display language of game ¹³ Default: en
te	UInt16	No	Time elapsed for game health reminder in seconds for operator (Reality check ¹⁴)
ri	UInt16	No	Reminder interval for game health reminder in seconds (Reality check ¹⁴)

¹² Please refer to [Bet Types](#) section for complete list

¹³ Please refer to [Languages](#) section for complete list

¹⁴ Please refer to [Reality Check](#) section for more information

op	varchar(200)	No	Custom parameter for operator, PG SOFT API will include parameter value upon verifying operator's player session Note: Please encode the value with UriEncode to avoid unexpected errors
f	varchar(200)	No	Game exit URL Default: Redirect to PG SOFT quit page Note: • Please encode the value with UriEncode to avoid unexpected errors • Assign value to <i>PGGameCloseUrl</i> to close the game window
oc	UInt8	No	Screen orientation check ¹⁵ when launching the game 0: Skip screen orientation check 1: Normal mode (Default)

extra_args Parameters (Web Lobby):

Parameter name	Data Type	Mandatory	Description
ops	varchar(200)	Yes	Token generated by operator system Note: • Please encode the value with UriEncode to avoid unexpected errors • Max 200 characters • Please set the parameter value as GUID format or any random string
l	varchar(50)	No	Display language ¹⁶ of web lobby Default: en
width	Integer	No	Page width of web lobby Default: Responsive Value range: Value ≥ 930

extra_args Parameters (Tournament):

Parameter name	Data Type	Mandatory	Description
ops	varchar(200)	Yes	Token generated by operator system Note:

¹⁵ Please refer to screenshot

¹⁶ Please refer to [Languages](#) section for complete list

			• Please encode the value with UrlEncode to avoid unexpected errors • Max 200 characters • Please set the parameter value as GUID format or any random string
l	varchar(50)	No	Display language ¹⁷ of web lobby Default: en

extra_args Parameters (Bet history interpreter¹⁸) :

Parameter name	Data Type	Mandatory	Description
t	String	Yes	Operator session
psid	String	Yes	Parent Bet ID
sid	String	Yes	Bet ID
gid	String	No	Game ID
lang	String	No	Language: • en (default) • zh
type	String	Yes	Fixed value: operator

Example:

Body:

```
operator_token=xxxx&path=%2F1%2Findex.html&extra_args=btt%3D1%26ops%3D12345-abcd-1234-abcd-12345&url_type=game-entry&client_ip=1.1.1.1
```

Response

HTTP Body response:

Response will be JSON data.

*** Note: The JSON response format is for reference only and might be updated accordingly.**

Successful response:

Parameter Name	Data Type	Description
error	object	Returns error message when an error has occurred. Value will be null if no error occurs.
data	string	Obfuscated output content.
version	string	The current version is 2 and may change with future upgrades.

¹⁷ Please refer to [Languages](#) section for complete list

¹⁸ Please refer to [Bet History Interpreter](#) section for complete information

Failed response:

Parameter Name	Data Type	Description
code	int	This is an error code. The error code is consistent with the HTTP status code. • 400: Request parameter error • 500: Internal Server Error
message	string	Detailed description of the error
tid	string	Trace ID is an Unique identity (GUID) of the request. Add ?trace_id=xxx in the query string to return the content of the trace id. Note: Please set the parameter value as GUID format

HTTP status code response:

Status Code	Description	Reason	Solution
200	Successful JSON Body response	-	Note: You need to check the error field in the JSON content. If there is a value in this field, an error has occurred.
504	Gateway Timeout	The request failed to reach the destination server successfully and a proxy server failed.	• Retry • Inform PG SOFT

Example for JSON Response (For reference only):

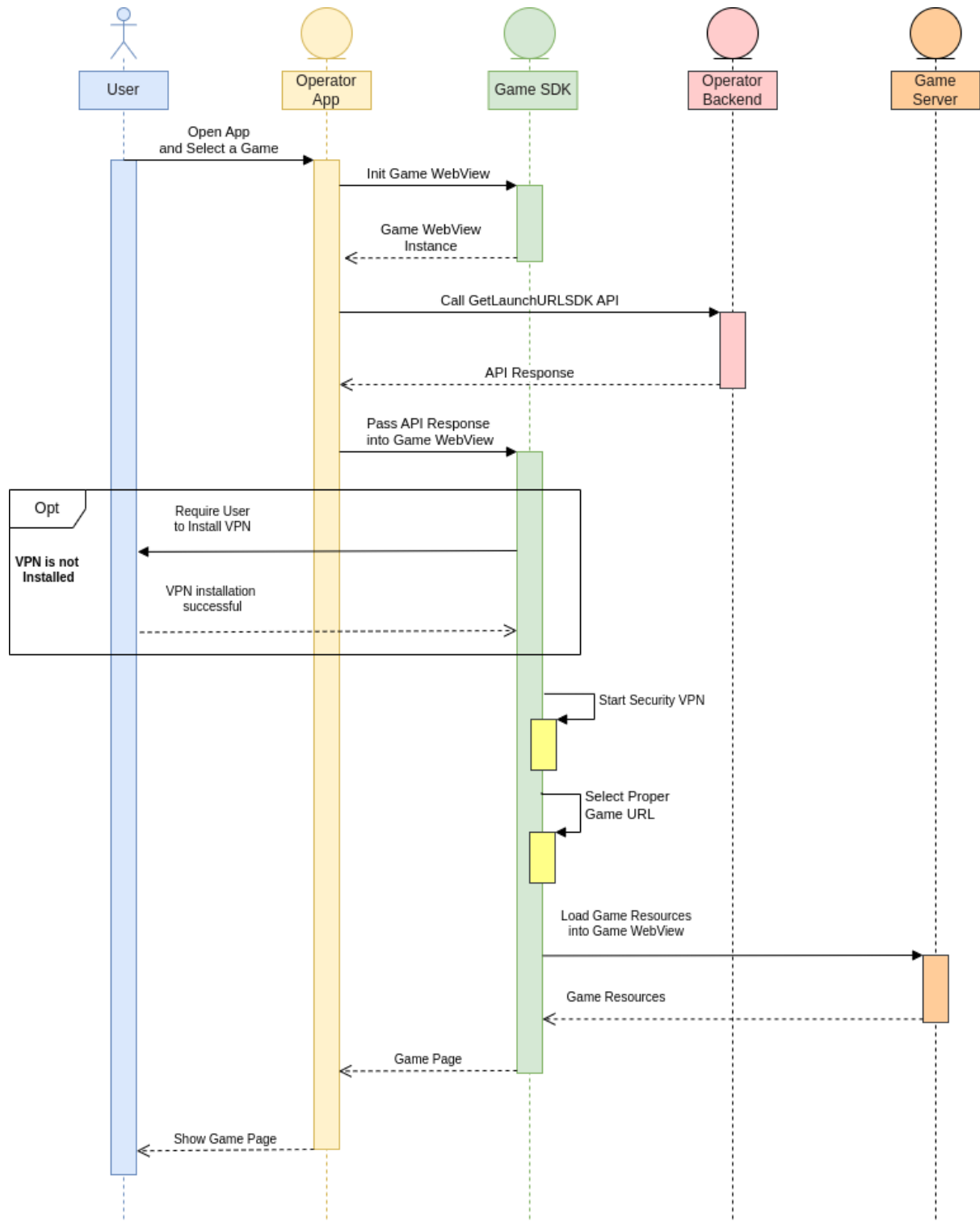
Successful response:

```
{
  "error": null,
  "version": "2",
  "data": "CAkQxzAD0odEi0OFisAR1kpRxwAHBkAZUZcMl0NGV8BEQJFDxZxChwyXAICHwAd..."
}
```

Failed response:

```
{
  "error": {
    "code": 400,
    "message": "Request validate error:\n\n\tclient_ip parse fail:
    2400:9800:1f1:4039:1:0:3866",
    "tid": "xxx9988"
  }
}
```

3.1.3 iOS SDK Scheme



3.1.3.1 Implementation

STEP 1: Prerequisites

1. Register an Apple account as a member of the Apple Developer Program.
2. Install the latest version of Xcode (15.0.1)
3. Obtain PG SDK XCFramework from PG SOFT

Note:

Staging SDK file: <https://app.pg-staging.com/game-sdk-ios/game-sdk.zip>

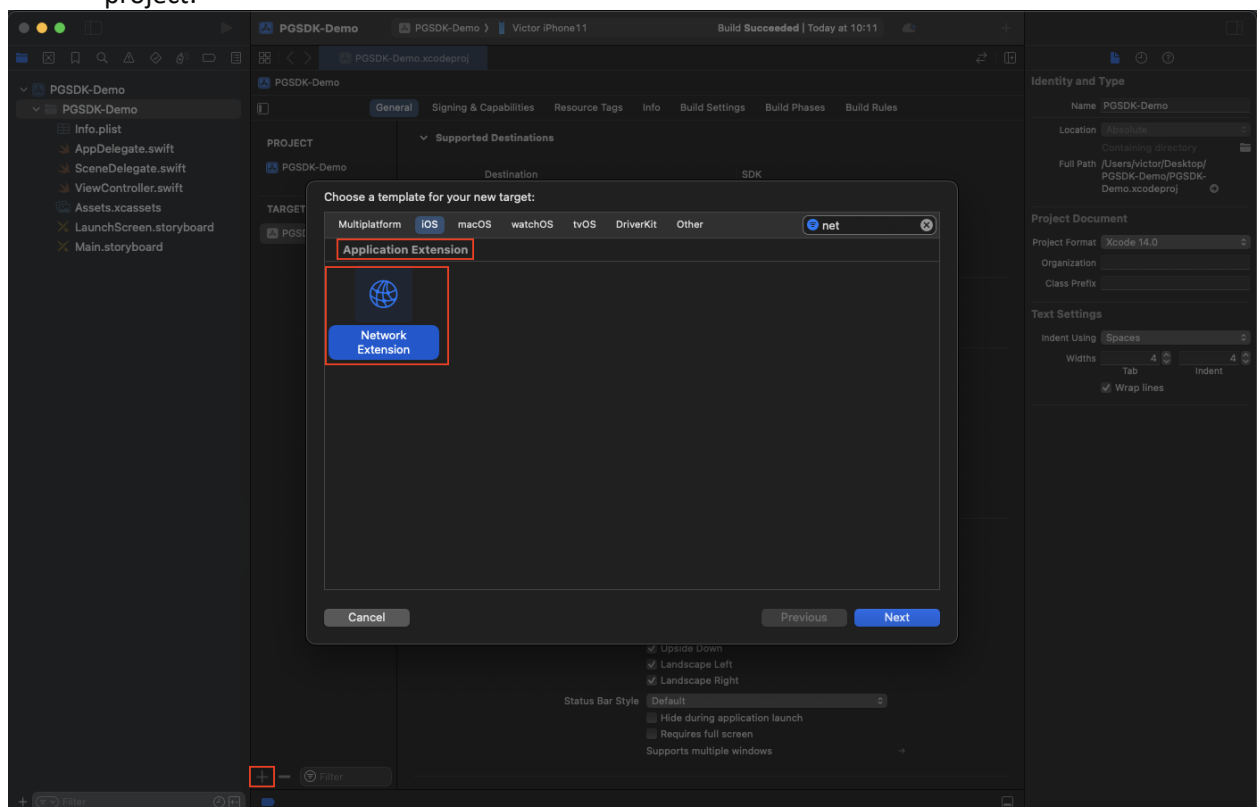
* Note: Production SDK file please request from PG SOFT support.

STEP 2: PG SDK integration steps

Add a Network Extension

Add a "Network Extension" target (Application Extension)

1. Click the "+" button to add the Application Extension's "Network Extension" to your iOS project:



2. Enter the “Product Name” in the pop-up window (for example: XXX-Tunnel) and select the “Packet Tunnel” option in the “Provider Type” drop-down option:

Choose options for your new target:

Product Name: PGSDK-Tunnel

Team: [Redacted] ▾

Organization Identifier: [Redacted]

Bundle Identifier: [Redacted].PGSDK-Tu...

Language: Swift ▾

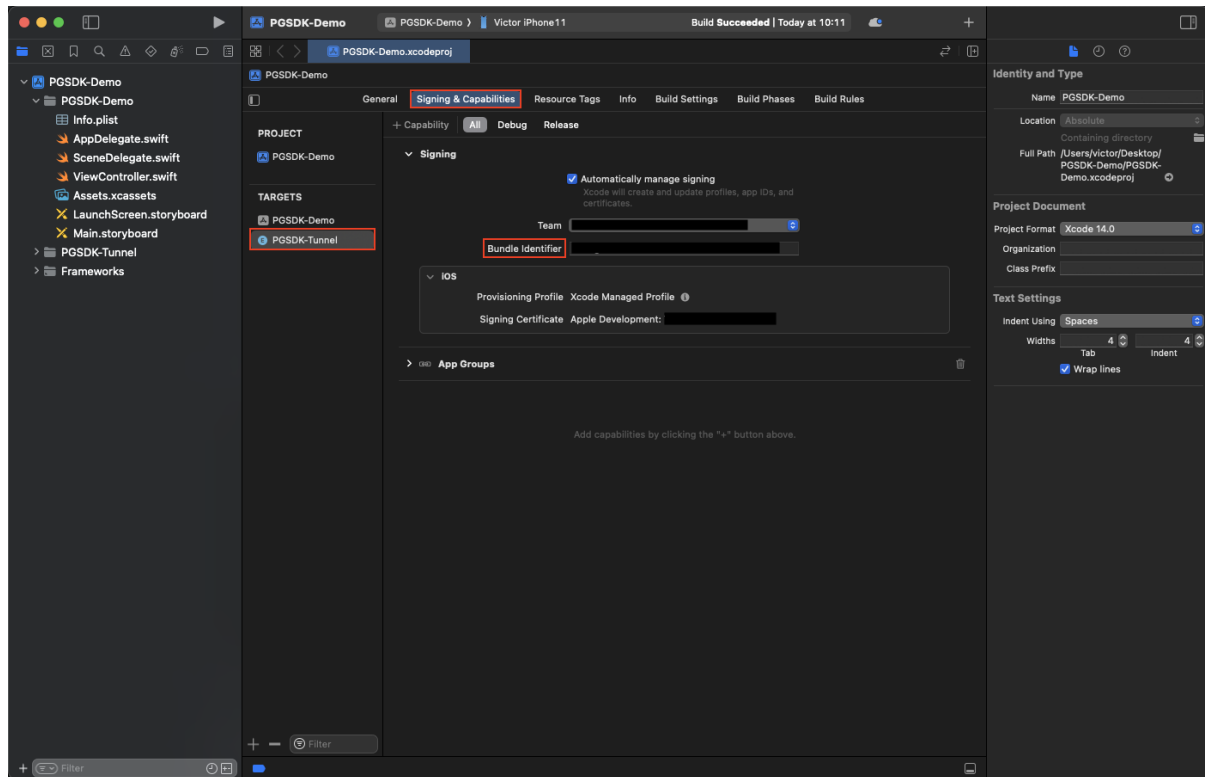
Provider Type: Packet Tunnel ▾

Project: PGSDK-Demo ▾

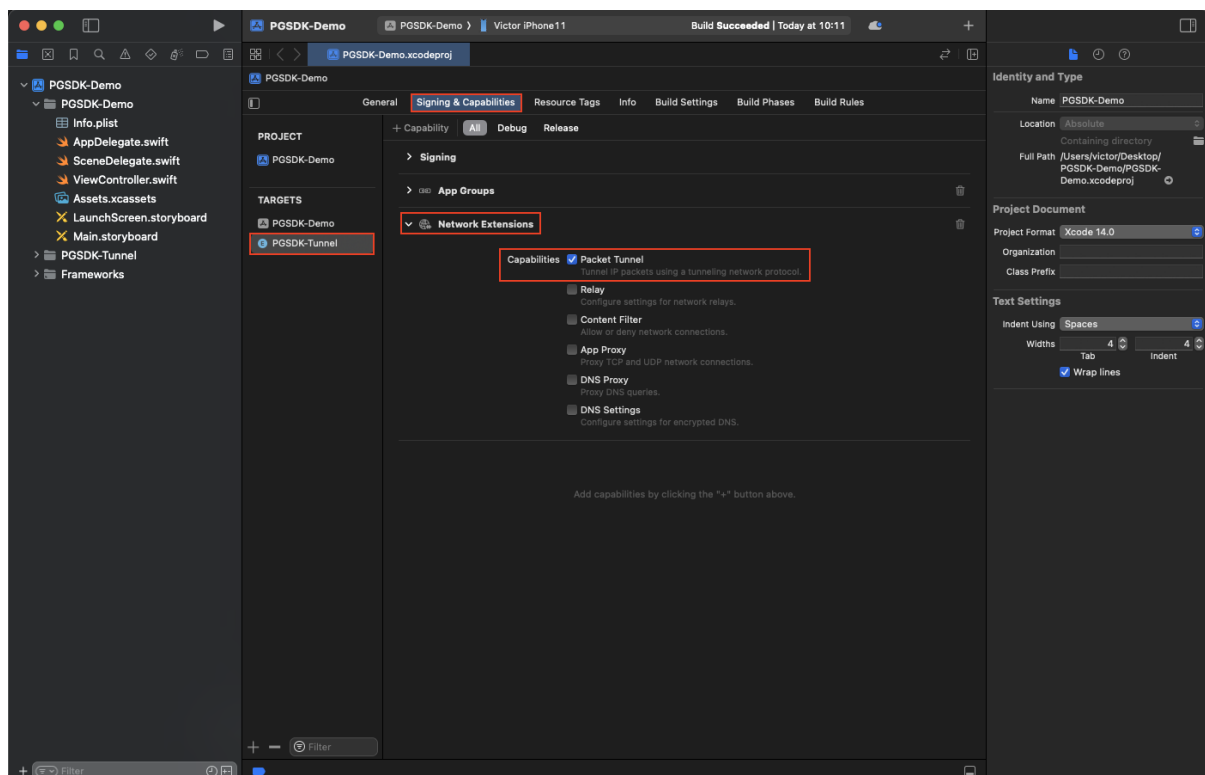
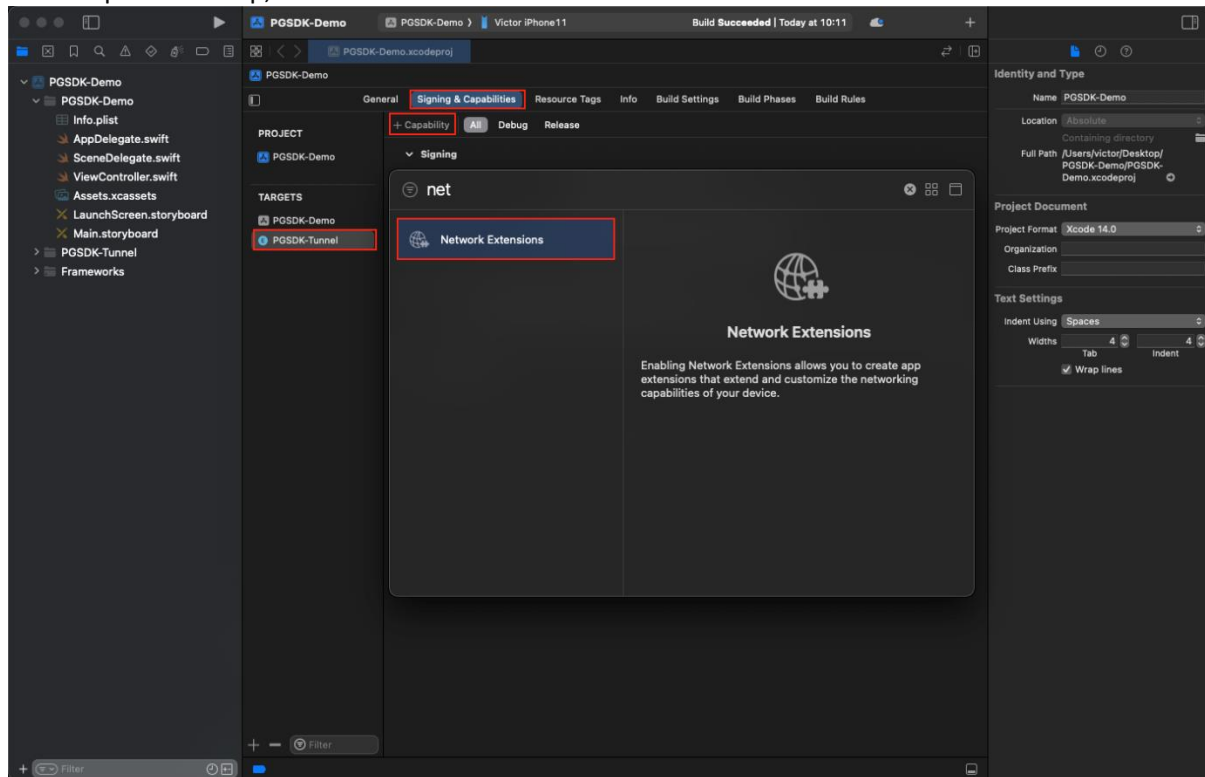
Embed in Application: PGSDK-Demo ▾

Cancel Previous Finish

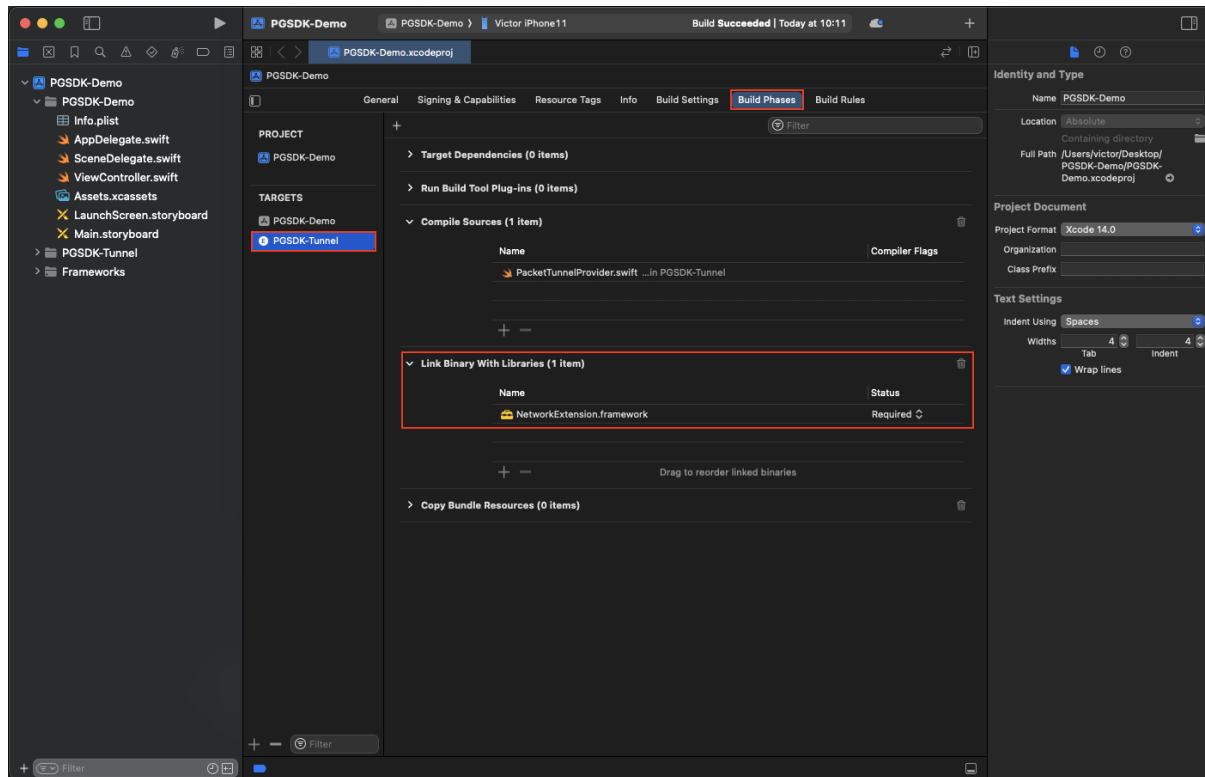
3. Please note that the “**Bundle Identifier**” of the Application Extension will be used later:



4. Add the “**Network Extensions**” capability to the “**Application Extension**” created in the previous step, and choose “**Packet Tunnel**” in the “**Network Extensions**” section:

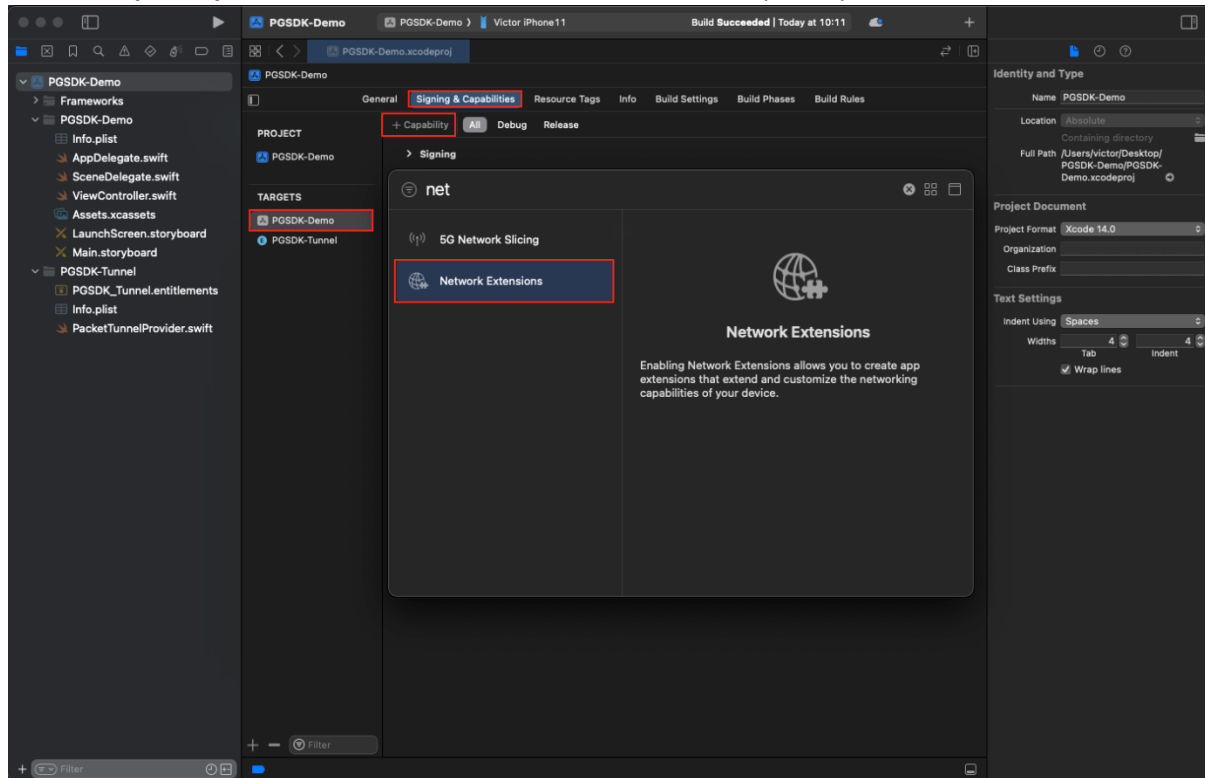


5. Check the framework configuration in the **"Build Phases"** tab of the **"Application Extension"**:

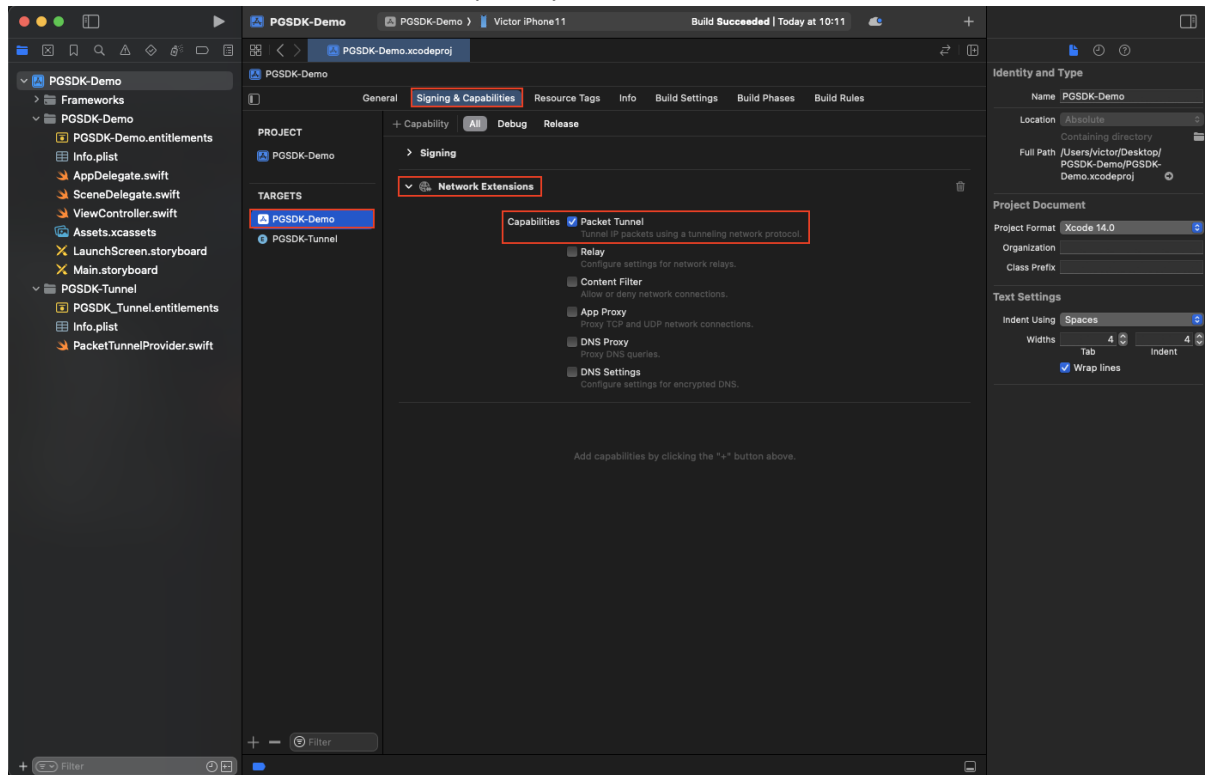


Add "Network Extensions" functionality to the main target

1. Select your primary **"target"** and go to the **"Signing & Capabilities"** tab. Click the **"+ Capability"** button to add the **"Network Extensions"** capability:

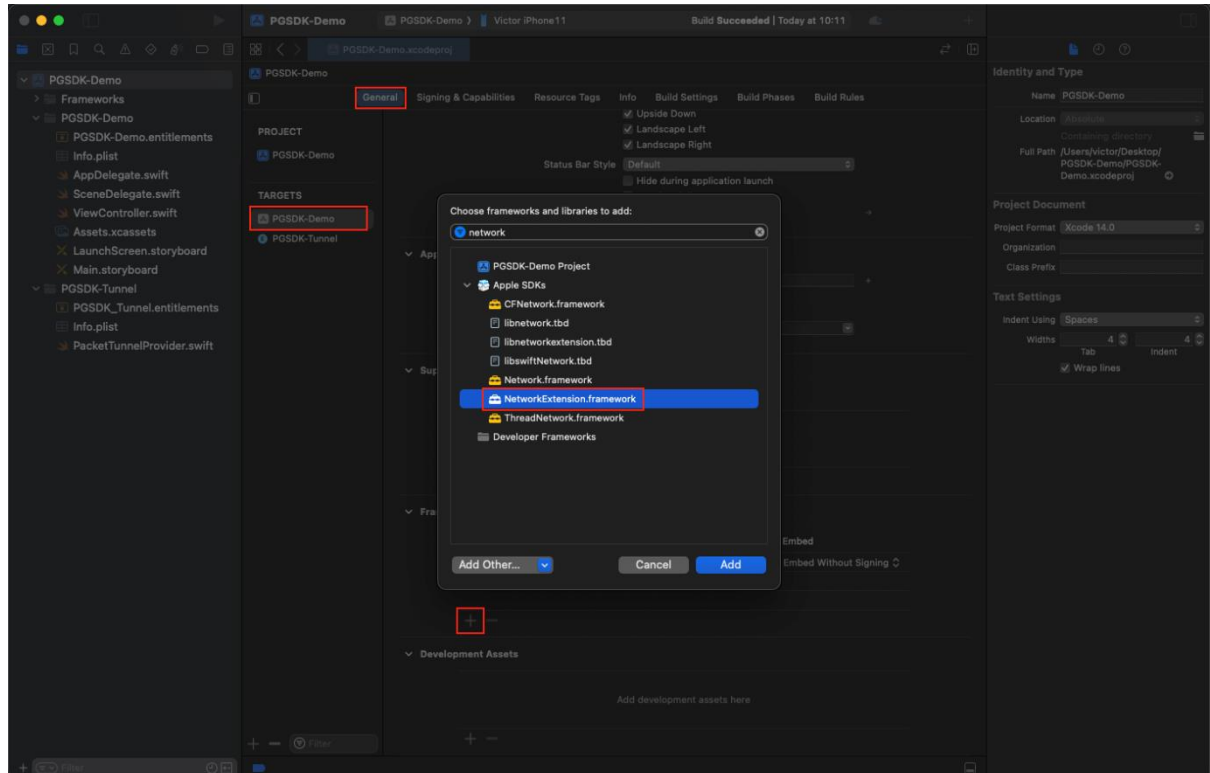


2. Choose the "Packet Tunnel" capability under the "Network Extensions" section:

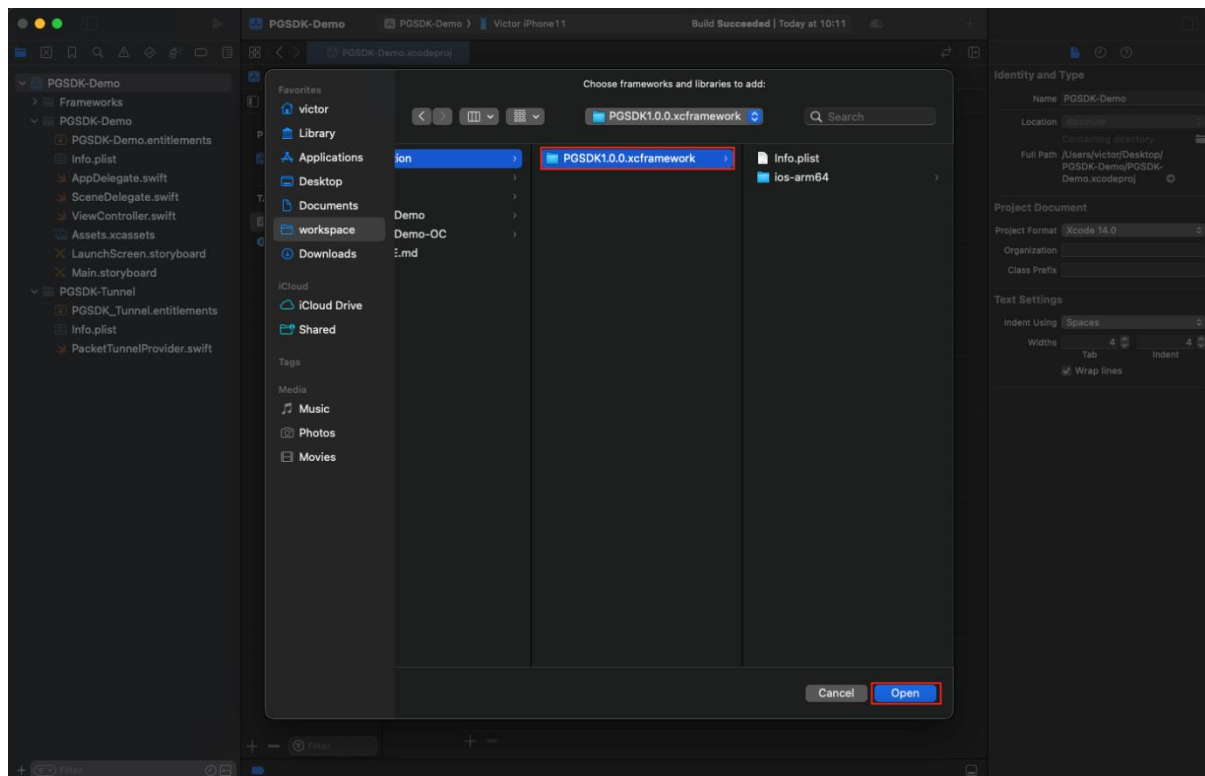
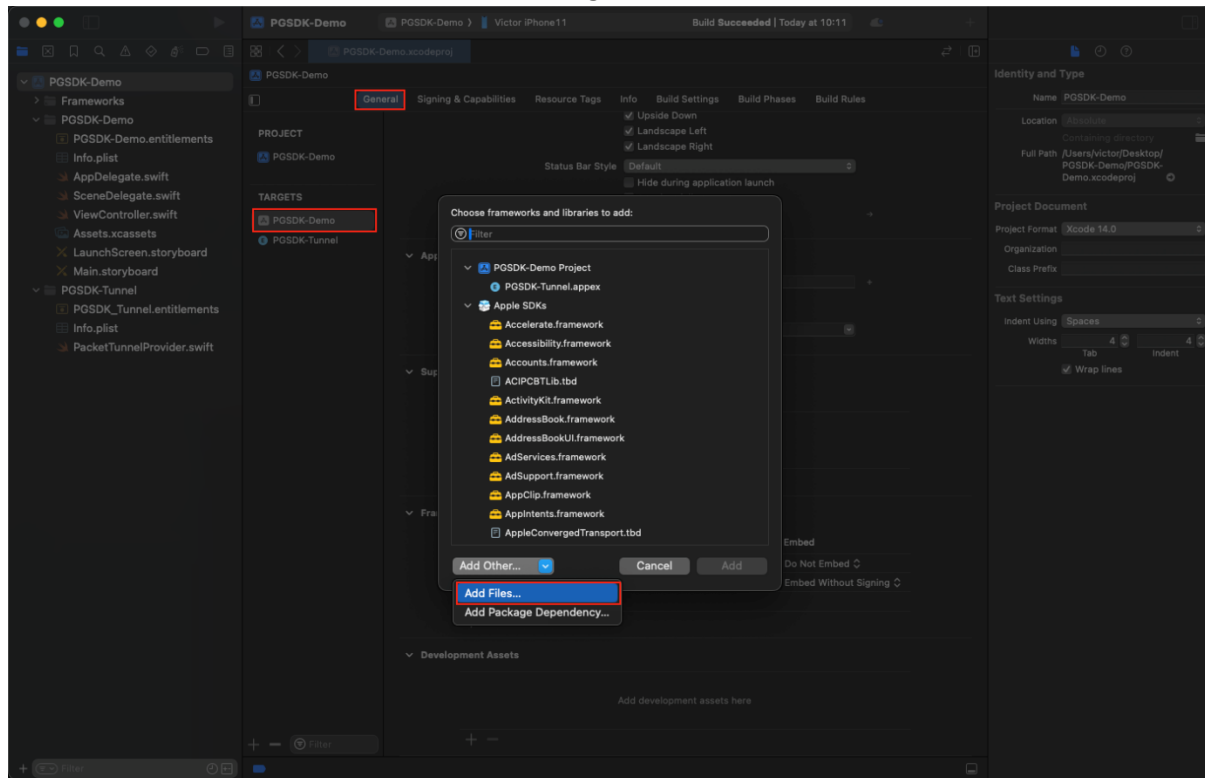


Add framework to "target"

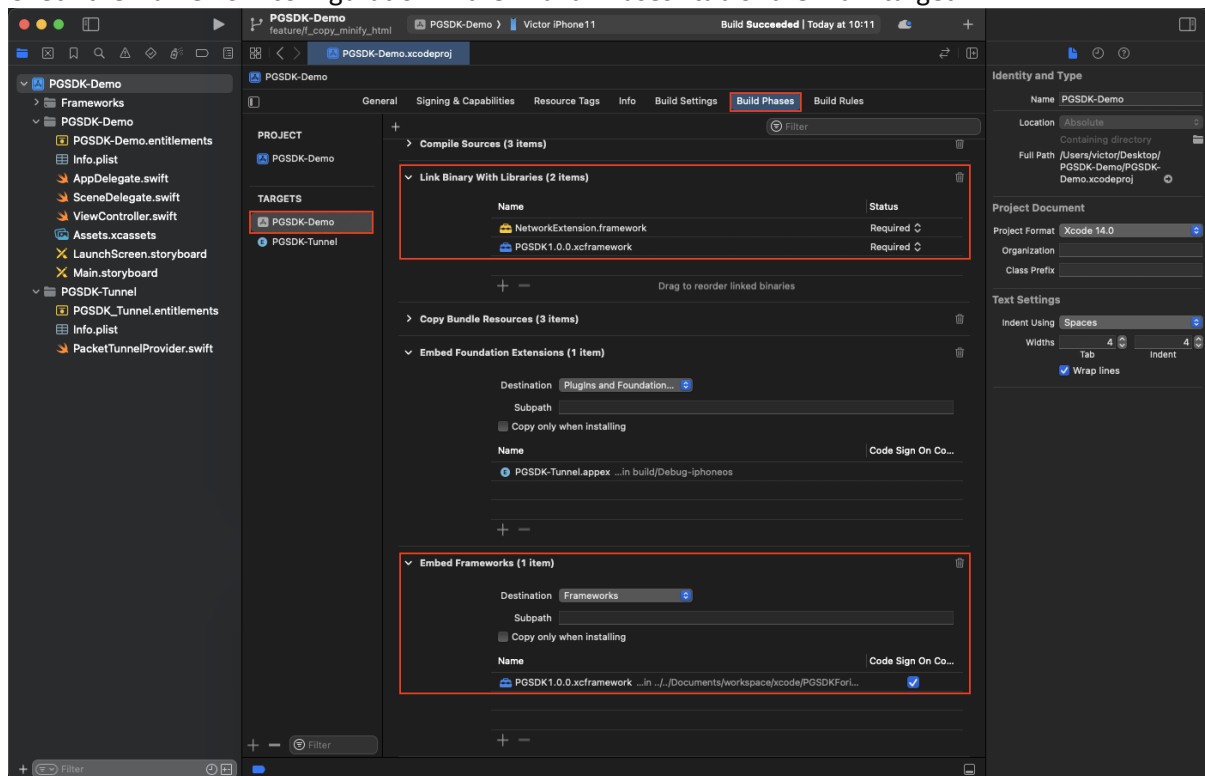
Add "NetworkExtension.framework" to the main target



Add "PGSDK[x.x.x].xcframework" to the main target



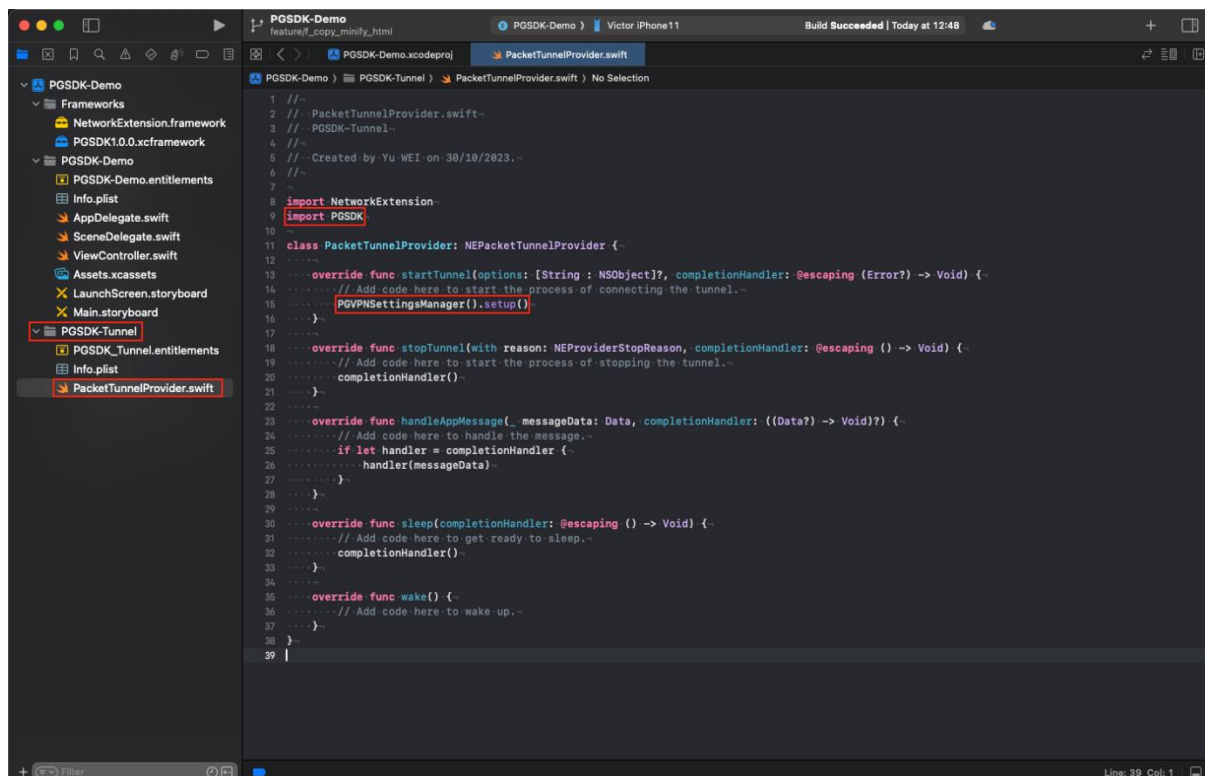
Check the framework configuration in the “Build Phases” tab of the main target



Integrate PacketTunnelProvider of PGSDK in Application Extension

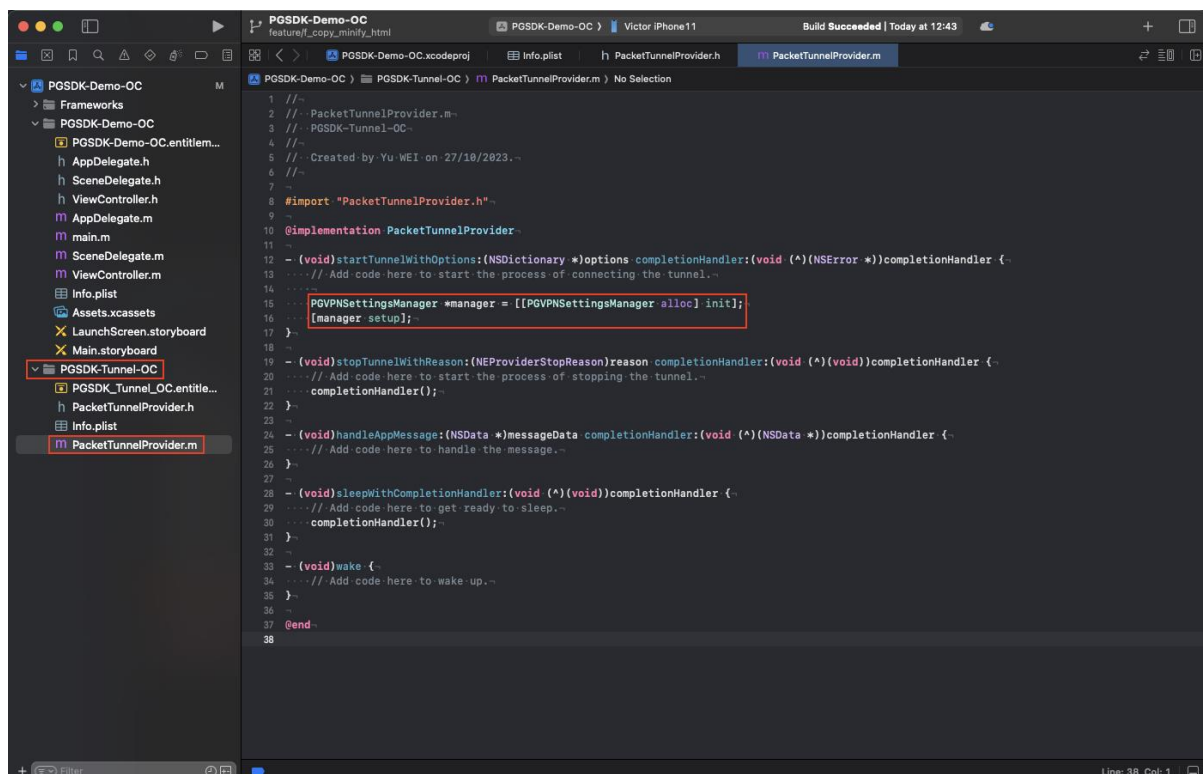
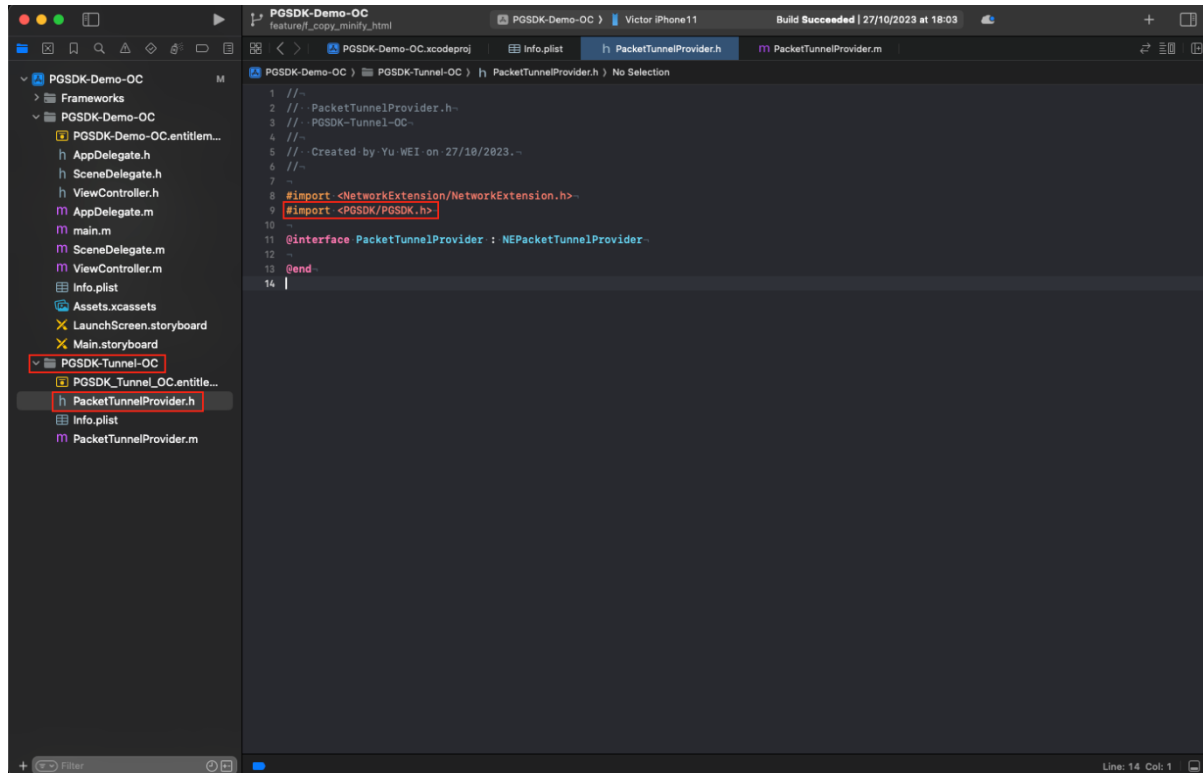
1. Open "**PacketTunnelProvider.swift**" (PacketTunnelProvider.h and PacketTunnelProvider.m in the Objective-C version) in "**Application Extension**"
2. Import "**PGSDK**" and add the following code to the "**startTunnel**" function:
Swift:

```
PGVPNSSettingsManager().setup()
```



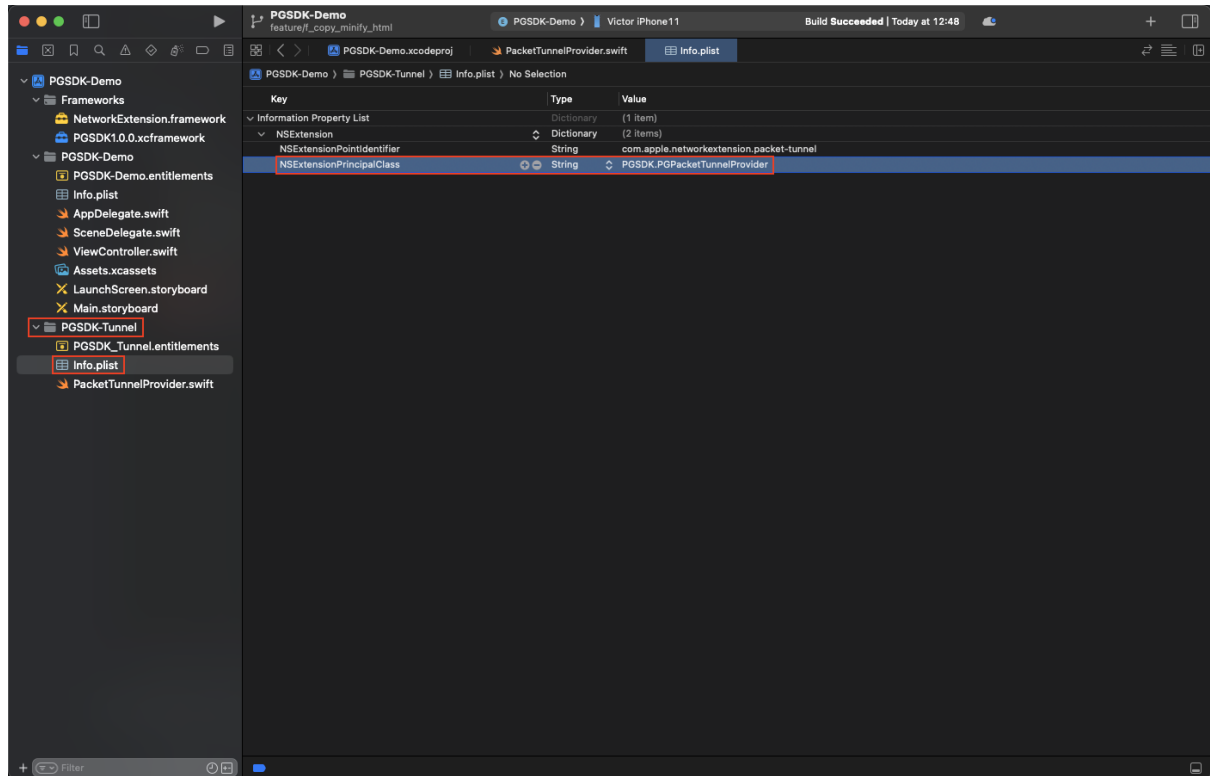
Objective-C:

```
PGVPNSettingsManager *manager = [[PGVPNSettingsManager alloc] init];
[manager setup];
```



3. Configure "NSExtensionPrincipalClass" in "info.plist" in "Application Extension":

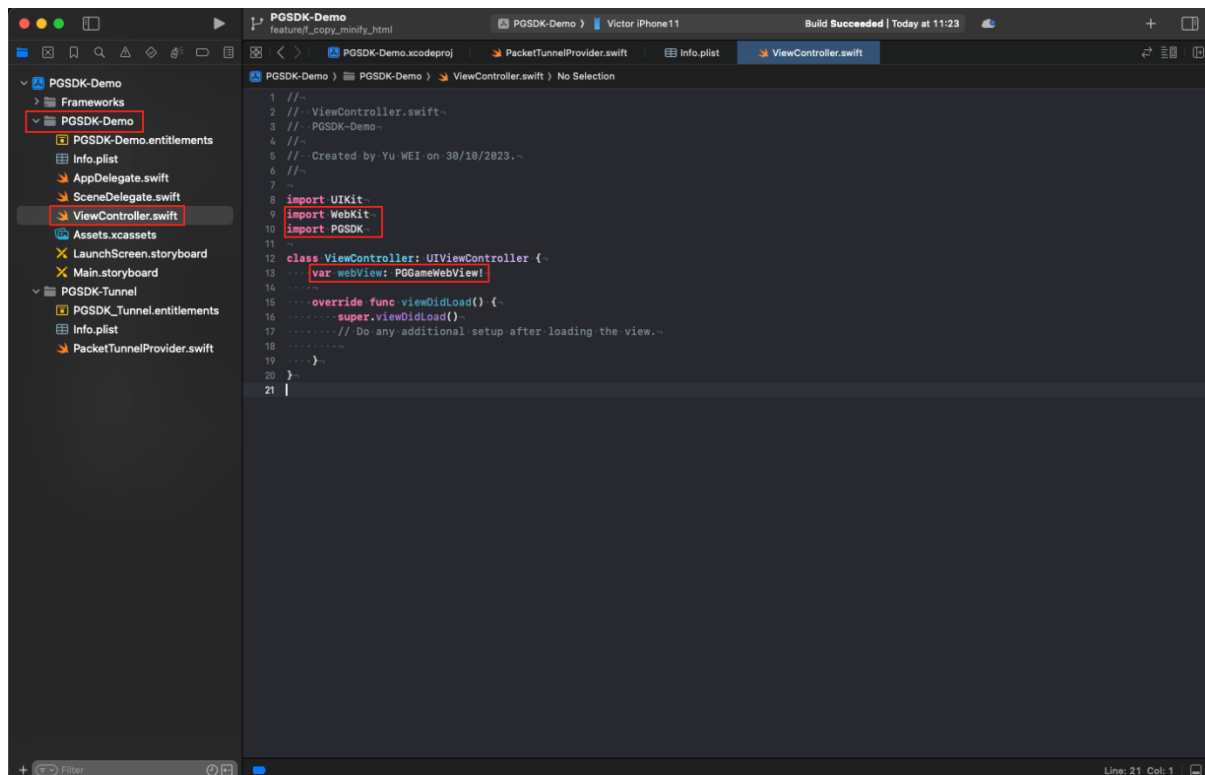
PGSDK.PGPacketTunnelProvider



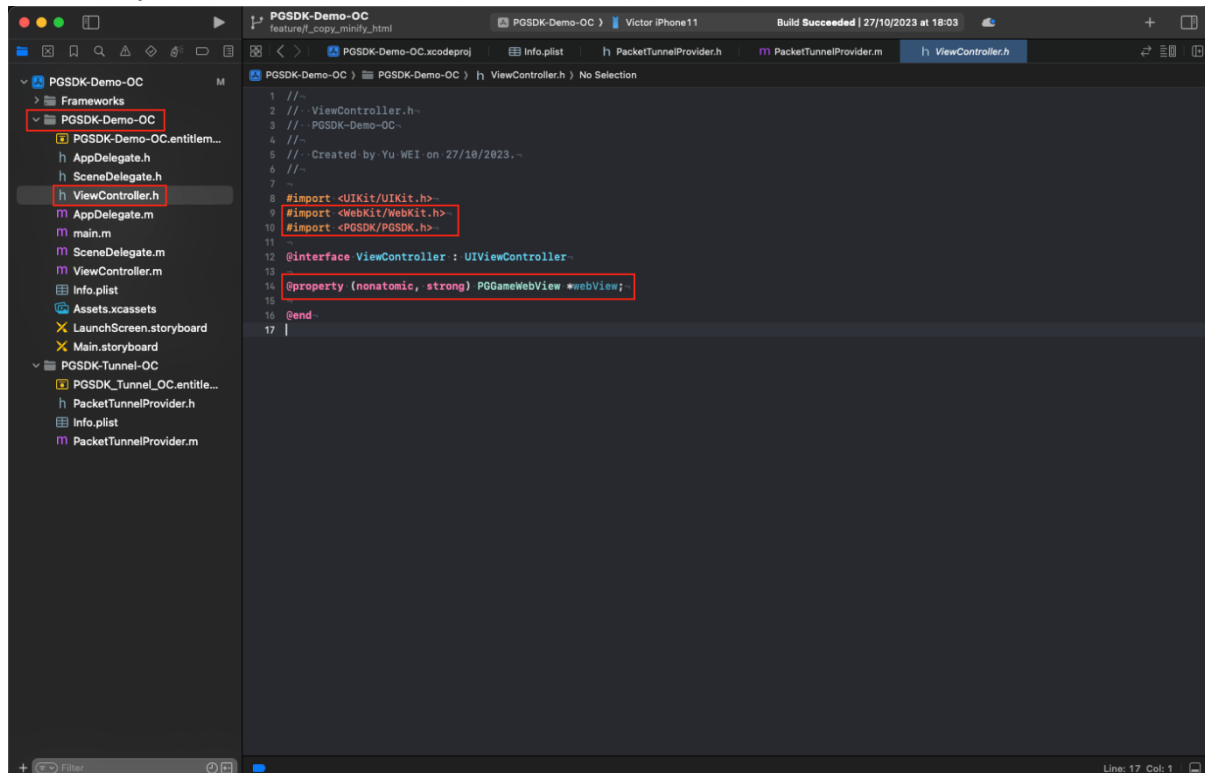
Integrate PGGameWebView of PGSDK in ViewController

1. Open "**ViewController.swift**" in the main target (ViewController.h and ViewController.m in the Objective-C version)
2. Import "**WebKit**" and "**PGSDK**" into "**ViewController**" and add a variable of type "**PGGameWebView**":

Swift:



Objective-C:



- Initialize "**PGGameWebView**" in the "**viewDidLoad**" function.
Please pass the value of section 2.1.1-3 (Bundle Identifier) to the "providerBundleIdentifier" parameter:
Please refer API Reference¹⁹ for "dnsType".
Swift:

```

self.webView = PGGameWebView(frame: self.view.bounds,
                             configuration: WKWebViewConfiguration(),
                             providerBundleIdentifier: "", // The value of
section 2.3.2.1.1-3
                             dnsType: PGDNSConfigType.google)
if let parentView = self.webView.superview {
    self.webView.frame = parentView.safeAreaLayoutGuide.layoutFrame
}
self.view.addSubview(self.webView)
  
```

¹⁹ Please refer to [API Reference](#) section for more information

Objective-C:

```
self.webView = [[PGGameWebView alloc] initWithFrame:self.view.bounds
configuration:[WKWebViewConfiguration
new]
providerBundleIdentifier:@""] // The value of
section 2.3.2.1.1-3
dnsType:PGDNSConfigTypeGoogle];
UIView *parentView = self.webView.superview;
if (parentView) {
    self.webView.frame = parentView.safeAreaLayoutGuide.layoutFrame;
}
[self.view addSubview:self.webView];
```

4. Call the "GetLaunchURLSDK" API²⁰ and load the response. Please call the "load" function of the PGGameWebView instance and pass the API response as a string:

Swift:

```
let apiResp = "" // Response of GetLaunchURLSDK API
self.webView.load(apiResp)
```

Objective-C:

```
NSString *apiResp = @""; // Response of GetLaunchURLSDK API
[self.webView load:apiResp];
```

²⁰ Please refer to [GetLaunchURLSDK API](#) section for more information

5. The complete code of "ViewController" for your reference.

Swift:

```
import UIKit
import WebKit
import PGSDK

class ViewController: UIViewController {
    var webView: PGGameWebView!

    override func viewDidLoad() {
        super.viewDidLoad()
        // Do any additional setup after loading the view.

        self.webView = PGGameWebView(frame: self.view.bounds,
                                      configuration:
WKWebViewConfiguration(),
                                      providerBundleIdentifier: "", // The
value of section 2.3.2.1.1-3
                                      dnsType: PGDNSConfigType.google)
        if let parentView = self.webView.superview {
            self.webView.frame = parentView.safeAreaLayoutGuide.layoutFrame
        }
        self.view.addSubview(self.webView)

        let apiResp = "" // Response of GetLaunchURLSDK API
        self.webView.load(apiResp)
    }
}
```

Objective-C:

```
#import <UIKit/UIKit.h>
#import <WebKit/WebKit.h>
#import <PGSDK/PGSDK.h>

@interface ViewController : UIViewController

@property (nonatomic, strong) PGGameWebView *webView;

@end
```

```
#import "ViewController.h"

@interface ViewController ()

@end

@implementation ViewController

- (void)viewDidLoad {
    [super viewDidLoad];
    // Do any additional setup after loading the view.

    self.webView = [[PGGameWebView alloc] initWithFrame:self.view.bounds
configuration:[WKWebViewConfiguration new]
providerBundleIdentifier:@""] // The value of
section 2.3.2.1.1-3
dnsType:PGDNSConfigTypeGoogle];
    UIView *parentView = self.webView.superview;
    if (parentView) {
        self.webView.frame = parentView.safeAreaLayoutGuide.layoutFrame;
    }
    [self.view addSubview:self.webView];

    NSString *apiResp = @""; // Response of GetLaunchURLSDK API
    [self.webView load:apiResp];
}

@end
```

3.1.4 API Reference

3.1.4.1 Implementation

PGDNSConfig

```
/**
 * Initialize a PGDNSConfig instance
 *
 * PGSDK provides custom DNS functionality, passing necessary parameters to
 * create PGDNSConfig instance
 *
 * @param {String} dnsName - DNS name (e.g. Google, Cloudflare)
 * @param {String} serverURL - DNS over HTTPS resolver URL
 * @param {String} apiURL - DNS over HTTPS JSON API URL
 * @param {[String]?} servers - Optional, DNS server IP (IPv4 or IPv6)
 * @returns {PGDNSConfig} PGDNSConfig instance
 */
init(dnsName: String, serverURL: String, apiURL: String, servers: [String]?)
```

PGDNSConfigType

```
/**
 * Enumeration type of DNS Type
 *
 * PGSDK provides 4 preconfigured DNS configuration (Google, Cloudflare,
 * Tencent, Alibaba). You can also use a "custom" type, in which case you need to
 * initialize a PGDNSConfig instance yourself
 */
enum PGDNSConfigType {
  case google
  case cloudflare
  case tencent
  case ali
  case custom
}
```

PGSDKDevToolKit

```
/**
 * Get PGSDK version string
 */
static func getVersionString() -> String?
```

PGGameWebView

```
/**
 * Initialize a PGGameWebView instance
 *
 * Initialize the PGGameWebView instance and use the preconfigured DNS
 configuration
 *
 * @param {CGRect} frame - Foundation CGRect
 * @param {WKWebViewConfiguration} configuration - WebKit
 WKWebViewConfiguration
 * @param {String} providerBundleIdentifier - Application Extension Bundle
 Identifier
 * @param {PGDNSConfigType} dnsType - Enumeration PGDNSConfigType
 * @returns {PGGameWebView} PGGameWebView instance
 */
init(frame: CGRect, configuration: WKWebViewConfiguration,
providerBundleIdentifier: String, dnsType: PGSDK.PGDNSConfigType)
```

```
/**
 * Initialize a PGGameWebView instance
 *
 * Initialize the PGGameWebView instance and use custom DNS configuration
 *
 * @param {CGRect} frame - Foundation CGRect
 * @param {WKWebViewConfiguration} configuration - WebKit
 WKWebViewConfiguration
 * @param {String} providerBundleIdentifier - Application Extension Bundle
 Identifier
 * @param {PGDNSConfig} dnsConfig - PGDNSConfig instance
 * @returns {PGGameWebView} PGGameWebView instance
 */
init(frame: CGRect, configuration: WKWebViewConfiguration,
providerBundleIdentifier: String, dnsConfig: PGSDK.PGDNSConfig)
```

```
/**
 * Load the game based on the "GetLaunchURLSDK" API response
 *
 * After initializing the PGGameWebView instance and getting the response of
 the "GetLaunchURLSDK" API, pass the response as a string to the "load"
 function of the PGGameWebView instance to load the game
 *
 * @param {String} responseString - API response
 */
func load(_ responseString: String)
```

PGVPNSettingsManager

```
/**  
 * Initialize a PGVPNSettingsManager instance  
 */  
init()
```

```
/**  
 * Perform necessary settings  
 */  
func setup()
```

3.1.5 Others

1. The minimum iOS version supported by PGSDK is 14.0.
2. Since this project involves the "Network Extension" framework, the iOS simulator does not support it, please run/debug on a physical device.

3.2 Sample Request

3.2.1 HTML Direct Launch Game Mode

Below is an example of HTML code for operator front-end. Player will launch PG SOFT games by clicking the launch game link.

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
</head>
<body><a href="https://<Operator Backend API>/pg-game-launcher-html-
mode.php?param1=xxx&param2=yyy" target="_blank">Launch Game</a></body>
</html>
```

1. Players will visit the operator's website and click on the PG SOFT game. The operator's frontend will call the backend service to obtain the HTML code.
2. In response to the frontend's request, the backend service will call the PG SOFT GetLaunchURLHTML API.
3. PG SOFT GetLaunchURLHTML API returns the HTML code.
4. The operator's backend API will pass the HTTP headers and HTML code from PG SOFT to the operator's frontend (which is the new window opened in STEP 1) without any modifications.
5. Upon receiving the HTML code, the frontend will execute it, displaying the HTML content on the player's browser (which is running the PG SOFT game).

3.2.2 iFrame Mode (Recommended)

Note:

Integrating iFrame mode helps effectively protect PG SOFT's domain. This integration method is recommended in regions where hijacking or domain blocking is common.

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head></head>
<body>
  <iframe allow="web-share *; clipboard-write *; screen-wake-lock *; fullscreen
*" src="https://<Operator Backend API>/pg-game-launcher-html-
mode.php?param1=xxx&param2=yyy">
  </iframe>
</body>
</html>
```

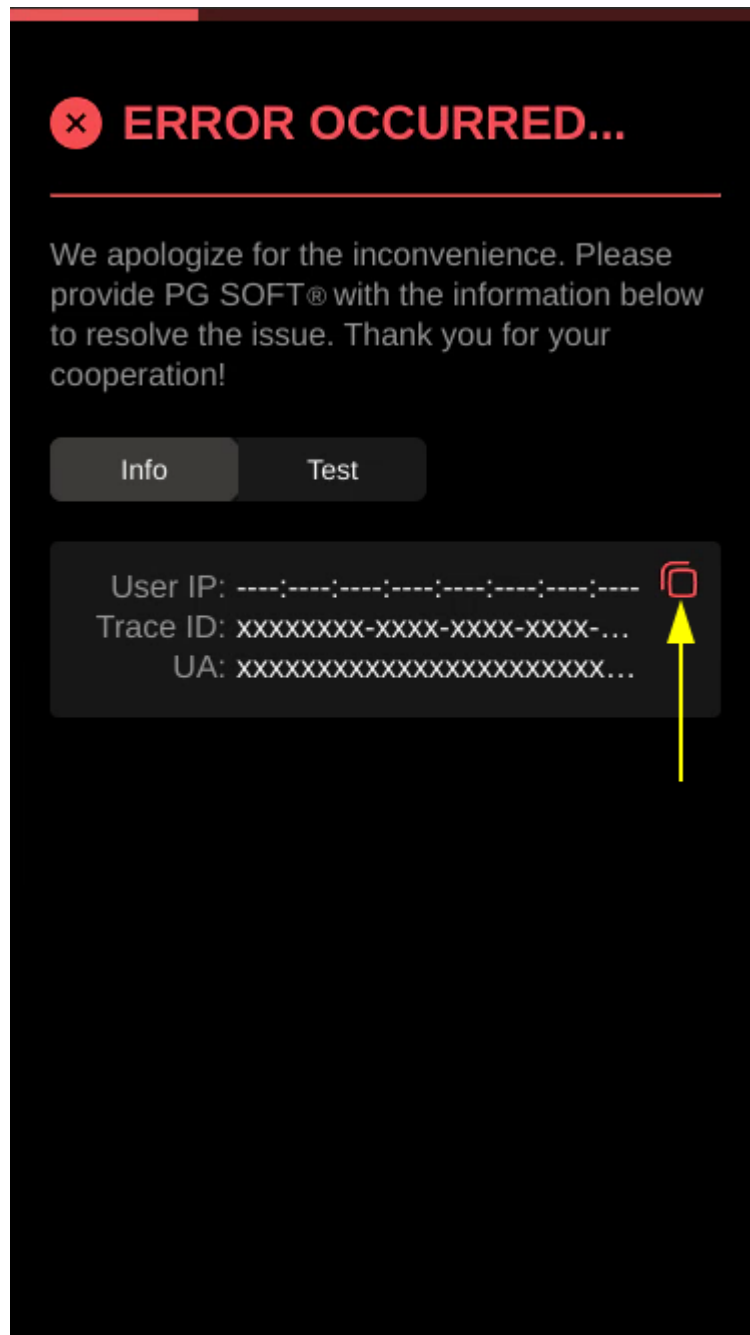
1. Players will visit the operator's website and click on the PG SOFT game. The operator's frontend will call the backend service to obtain the HTML code.
2. In response to the frontend's request, the backend service will call the PG SOFT GetLaunchURLHTML API.
3. PG SOFT GetLaunchURLHTML API returns the HTML code.
4. The operator's backend API will pass the HTTP headers and HTML code from PG SOFT to the operator's frontend (which is the new window opened in STEP 1) without any modifications.
5. Upon receiving the HTML code, the frontend will execute it, displaying the HTML content on the player's browser (which is running the PG SOFT game).

Note:

To ensure PG SOFT games launch properly, the allow attribute must be included in the iframe.

The reason is as follows:

- Clipboard-write:



When players experience network issues in the game launcher, they can click the Copy button and provide the copied information to PG SOFT Support. This helps PG SOFT identify and resolve the issue more efficiently.



In the PG Verification, if a player is unable to open the PG SOFT verification page directly, they can still copy the URL and perform the verification manually.

- Fullscreen and screen-wake-lock:
Required by game engine.

3.2.3 Ajax Mode (Not Recommended)

* **Note:** Ajax mode is not recommended due to the following potential issues:

- **Scenario 1:** In iOS Safari, a "Pop-ups blocked" prompt may appear, preventing the game window from opening. Users must tap "Always Show" and then tap "Open" again to proceed. (This corresponds to Method 1 in the sample code below.)
- The browser's Back button will not return the user to the page they were on before launching the PG SOFT game. (This corresponds to Method 2 in the sample code below.)

Examples:

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.6.4/jquery.min.js"></script>
<script>
$(document).ready(function(){
    $("button").click(function(){
        $.getJSON("https://httpbin.org/anything", { html: '<h1>New
Window</h1><p>Hello World!</p>' }, function(data) {
            var html = data.args.html;

            // Method 1, use new window to execute HTML
            var newWindow = window.open('', '_blank');
            newWindow.document.open();
            newWindow.document.write(html);
            newWindow.document.close();

            // Method 2, use existing window to execute HTML.
            document.write(html);
        });
    });
});
</script>
</head>
<body><button>Launch Game</button></body>
</html>
```

1. Players will click the 'Launch Game' button. The operator's backend API will be triggered via Ajax.
2. Operator's backend API will send a request to PG SOFT GetLaunchURLHTML API.
3. PG SOFT GetLaunchURLHTML API returns the HTML code.
4. **The operator's backend API will pass the HTTP headers and HTML code from PG SOFT to the operator's frontend (which is the new window opened in STEP 1) without any modifications.**
5. Upon receiving the response from the operator's backend API, Ajax will execute the callback function, displaying the HTML content on the player's browser (which is running the PG SOFT game).

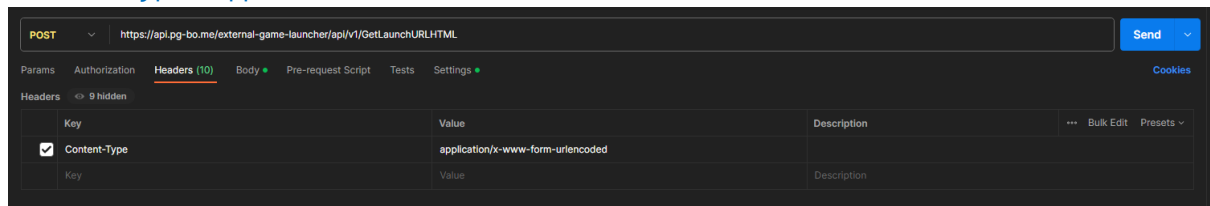
3.2.4 POSTMAN (RAW body)

The example below demonstrates how to call the PG SOFT GetLaunchURLHTML API using Postman to retrieve the HTML code.

Request:

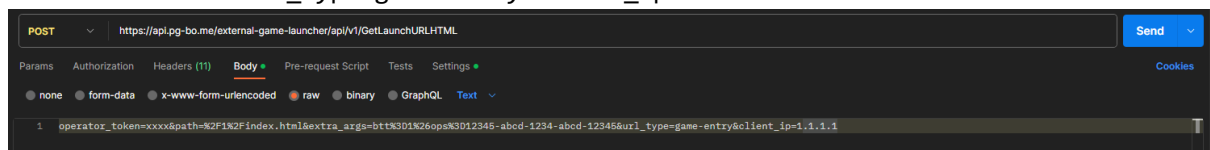
Headers:

Content-Type: application/x-www-form-urlencoded



Body (raw):

operator_token=xxxx&path=%2F1%2Findex.html&extra_args=btt%3D1%26ops%3D12345-abcd-1234-abcd-12345&url_type=game-entry&client_ip=1.1.1.1



3.2.5 CURL command

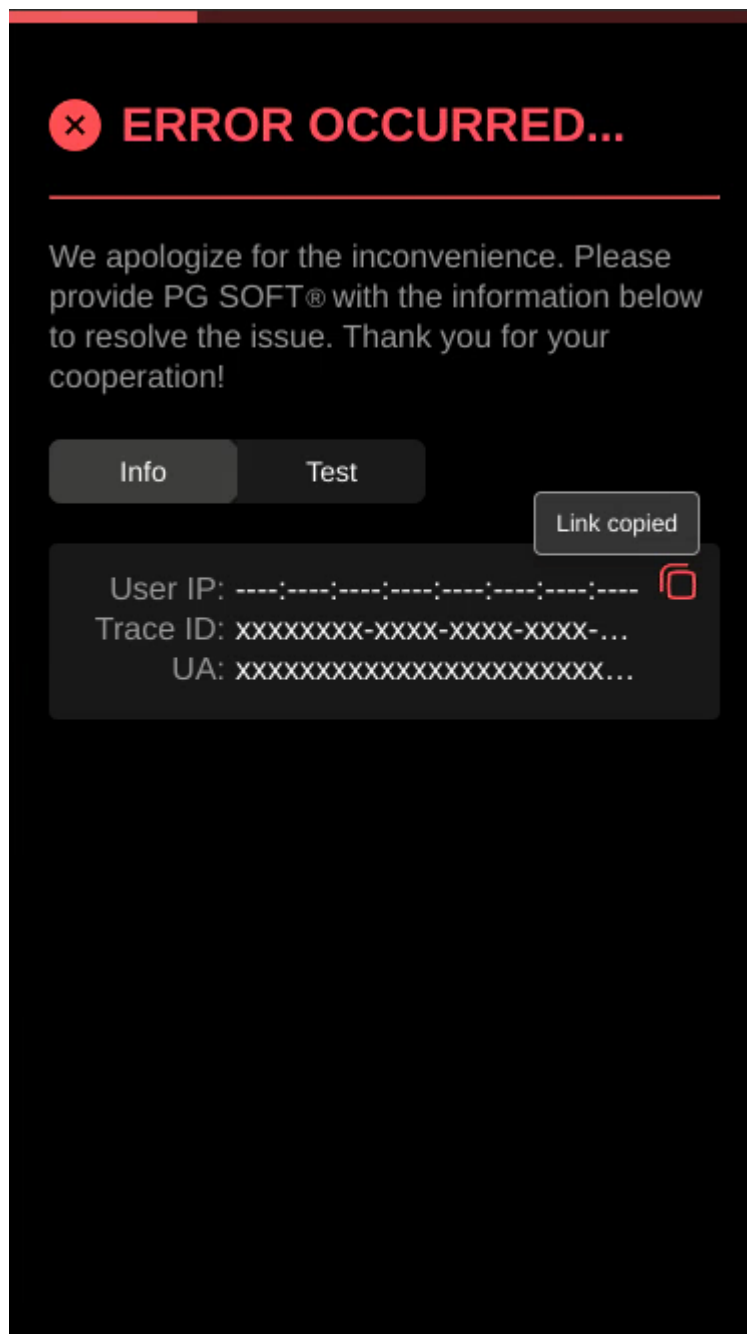
The example below demonstrates how to call the PG SOFT GetLaunchURLHTML API using a CURL command to retrieve the HTML code.

Request:

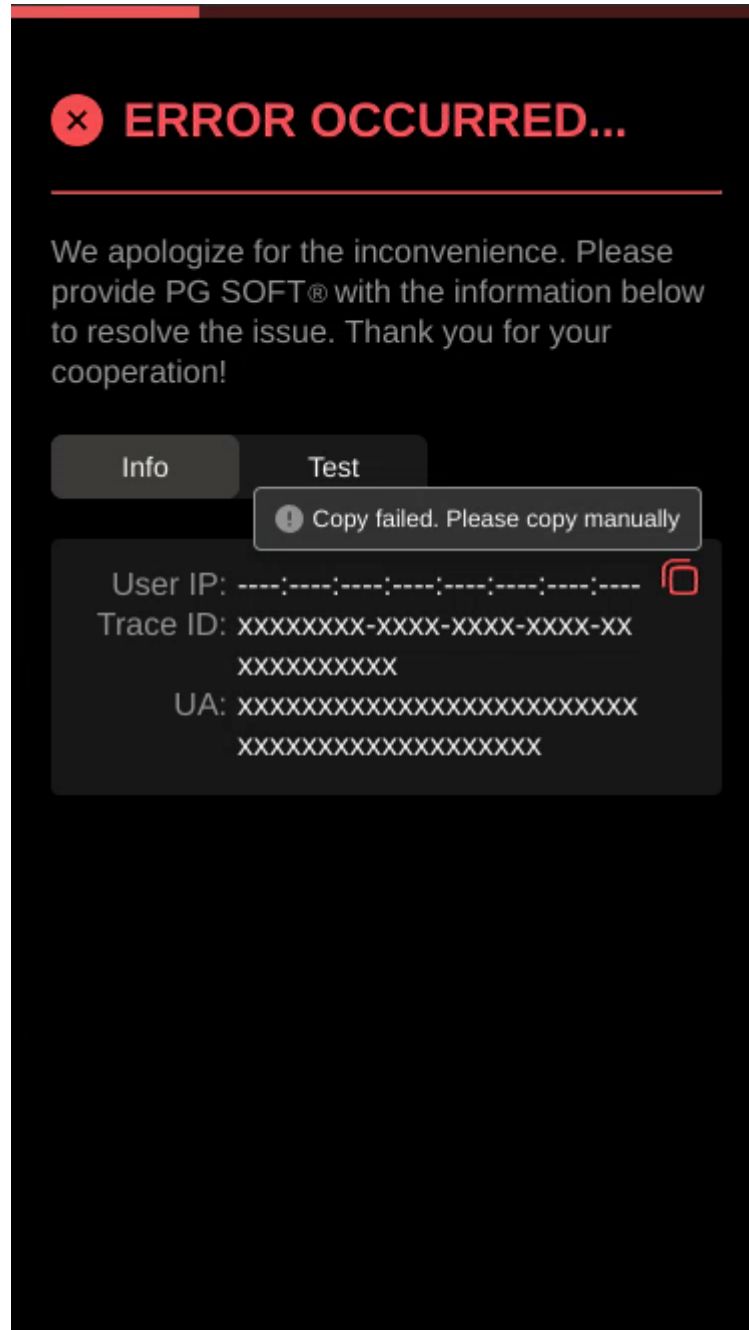
```
curl -v -X POST "https://api.pg-bo.me/external-game-launcher/api/v1/GetLaunchURLHTML"  
-H "Content-Type: application/x-www-form-urlencoded"  
-d  
"operator_token=xxxx&path=%2F1%2Findex.html&extra_args=btt%3D1%26ops%3D12345-abcd-1234-abcd-12345&url_type=game-entry&client_ip=1.1.1.1"
```

3.2.6 Troubleshooting Method

1. Copy the request information.
Example as below:



2. If the browser does not allow copying, the collapsed content will automatically expand. In that case, please provide a screenshot instead like below:



- To speed up troubleshooting, please provide a screenshot of each link under the Test tab:

ERROR OCCURRED...

We apologize for the inconvenience. Please provide PG SOFT® with the information below to resolve the issue. Thank you for your cooperation!

Info

Test

01
(000/000/000/000)

02
(000/000/000/000)

03
(000/000/000/000)

3.3 Game Launch Restrictions

Multiple windows or multiple game launches are unsupported by PG SOFT games. Players will be locked and kick out of games, and encounter errors when the below are detected:

S1302XXXXXXXX: Unsuccessful Game Authorization

S1309XXXXXXXX: Player account been suspended

S1305XXXXXXXX: Server request failure

When encountering the above errors, players may close the game windows and relaunch the game to unlock their player accounts:

S1302XXXXXXXX: Please relaunch the game

S1309XXXXXXXX: Player account will be unlocked automatically after 10 minutes, please relaunch the game after 10 minutes

S1305XXXXXXXX: Please relaunch the game

3.4 FAQ

1. 403 Error when calling the game-launcher API
Access to the game-launcher API requires API whitelisting. Please contact PG SOFT Support for API whitelisting.
2. DNS error, network error, or connection timeout
Please check the player's network settings.
3. HTTP Error 400/404
Please check the Content-Type of the request.
4. HTTP Error 500
Please retry the request. If the issue persists, contact PG SOFT for further support.
5. Invalid characters appearing on the page:
 - a. Ensure Gzip compression is enabled in the browser.
 - b. Check if the operator's server is passing the Content-Encoding HTTP header correctly.
6. Browser-side scripting errors
For supported browser types and versions, please contact PG SOFT Support.

4 Game Launch JS Library

Operator is required to launch the game by using PG SOFT JS library to provide the best gaming experience to players:

- Implementation of automatically adapting pop-up window according to the user's display size.
- PG SOFT games are all designed to be displayed vertically, the pop-up window format is consistent with product design.
- Achieving the optimal gaming Interface height will allow users to feel visually comfortable, in line with usage habits.
- Presenting the best in-game visual quality to the players, providing the best gaming experience.

4.1 Avoid Browser Pop-up Blockers

The general rules that popup blockers will engage are:

- Invoked from JavaScript that is not invoked by direct user action.
- Pop-up content are not from the same domain as the source.

Operator is recommended to follow the instructions below to avoid browser pop-up blockers:

- Create a frontend page and call their own backend service (HTML Scheme²¹).
- When players click the game launch button/link, use the PG SOFT JS function (PGSDK.launchGame) and launch the operator frontend page.

²¹ Please refer to [HTML Scheme](#) section for more information

4.2 Implementation

Step 1

Include the following line in the HTML <head> tags:

```
<script defer src="https://public.pg-staging.com/sdk/main.min.js"></script>
```

Note:

Operator can self-host the JS library as well: <https://public.pg-staging.com/sdk/main.min.js>

Step 2

Operator is required to create a self-host frontend page to call their own backend service (HTML Scheme²²) in order to obtain the HTML code for launching the game.

Sample code of operator's frontend page:

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.6.4/jquery.min.js">
</script>
<script>
    $(document).ready(function(){
        $.post( "https://<Operator backend service>/api",
            { param1: "value1", param2: "value2" },
            function(data) {
                window.document.write(data);//Execute HTML Code
            });
    });
</script>
</head>
</html>
```

²² Please refer to [HTML Scheme](#) section for more information

Step 3

Use the following JS script to launch PG SOFT games.

```
PGSDK.launchGame('{operator_frontend_URL}');
```

Parameters:

Parameter name	Data Type	Mandatory	Description
operator_frontend_URL	String	Yes	Operator self-host frontend page in Step 2, which will call their own backend service

Example:

```
PGSDK.launchGame('https://[OperatorDomain]/operator-frontend.html');
```

5 Integration API

5.1 Operator Provided API

5.1.1 API Formats

5.1.1.1 Request

The API uses HTTP form methods and a RESTful endpoint structure. PG SOFT will send requests in the following content type:

Content-Type: `application/x-www-form-urlencoded`

5.1.1.2 Response

API returns JSON-formatted responses. For successful and failed API request, operator is required to return an API response with HTTP status code 200.

The API response should be returned within 10 seconds to prevent the request being treated as a timeout request.

API uses global JSON response format for all API methods, with the following content type:

Content-Type: `application/json`

Parameters:

Parameter name	Data Type	Description
<code>data</code> ²³	JSON Object	Response information for API methods
<code>error</code>	JSON Object	Error information when an error or exception occur

Example:

Successful Response

```
{
  "data": {
    [API method response. JSON object format may vary depending on API methods]
  },
  "error": null,
}
```

Failed/Error Response

```
{
  "data": null,
  "error": {
    "code": "[Error code]",
    "message": "[Error message]"
  }
}
```

²³ This field will show *null* value in error response

5.1.1.3 Hash Authentication (Optional)

We highly recommend that the operator integrate hash authentication to secure every request. If the operator chooses to integrate with hash authentication, PG SOFT system will validate the hash information provided in the header fields for every request.

Note:

- Hash Authentication is disabled by default. Please request with PG SOFT if operator wishes to integrate Hash Authentication.
- The operator shall perform encryption and calculations based on the received request parameters and the body parameters should not be hardcoded.
- Hash header supports all APIs except for GetLaunchURLHTML and GetLaunchURLSDK.

5.1.1.3.1 Header Fields

The following table describes the various request headers in the preceding example:

Component	Description
Host	API domain Example: <code>apiexample.pgsoft.com</code>
x-date	Current date (UTC) in YYYYMMDD format Example: <code>20190902</code>
x-content-sha256	SHA256 hash of the request string body Example: <code>1700116101f424b9f6fc695b4dbaf2b7b0ee763ba1b3b53298e3069143ed46f1</code>
Authorization	Combination of various components for requesting authentication Example: <code>PWS-HMAC-SHA256 Credential=20190902/OPERATOROKENEXAMPLE/pws/v1, SignedHeaders=host;x-content-sha256;x-date, Signature=d78220cf06ae85f9d1db11dad9c3fd926799619eab3d28574aadb8cf328cd7aa</code>

Example

Host:
apiexample.pgsoft.com
x-date: 20190902
x-content-sha256:
1700116101f424b9f6fc695b4dbaf2b7b0ee763ba1b3b53298e3069143ed46f1
Authorization: PWS-HMAC-SHA256
Credential=20190902/OPERATORTOKENEXAMPLE/pws/v1,SignedHeaders=host;x-
-content-sha256;x-
date,Signature=d78220cf06ae85f9d1db11dad9c3fd926799619eab3d28574aadb
8cf328cd7aa

Header: x-content-sha256

Calculate a SHA256 hash of the request string body:

Example

POST body:
operator_token=abc123&secret_key=alb25cde5f3gh46ijkl&count=5000&bet_
type=1&row_version=1346592723000

x-content-sha256:
1700116101f424b9f6fc695b4dbaf2b7b0ee763ba1b3b53298e3069143ed46f1

Header: Authentication

The following table describes the various components of the Authorization header value in the preceding example:

Component	Description
Credential	Your access key ID and the scope information, which includes the date, operator token, and service that were used to calculate the signature. This string has the following form: <pre>{x-date}/{operator_token}/pws/v1</pre> Example: 20190902/OPERATOR_TOKEN_EXAMPLE/pws/v1
SignedHeaders	A semicolon-separated list of request headers that you will use to compute the Signature. The list includes header names only, and the header names must be in lowercase. Fixed Value: host;x-content-sha256;x-date
Signature	The 256-bit signature is expressed as 64 lowercase hexadecimal characters. Example: d78220cf06ae85f9d1db11dad9c3fd926799619eab3d28574aad db8cf328cd7aa

To calculate a signature, you first need a string that includes `{host}{x-content-sha256}{x-date}` to sign. You will then calculate a HMAC-SHA256 hash of the string to sign by using a signing key:

```
hmac-sha256({salt}, {host}{x-content-sha256}{x-date})
```

Calculate a HMAC-SHA256 hash of the string body:

Example

```
Salt           : SALT_EXAMPLE
Host           : apiexample.pgsoft.com
x-date         : 20190902
x-content-sha256 :
1700116101f424b9f6fc695b4dbaf2b7b0ee763ba1b3b53298e3069143ed46f1
```

Signature:

```
d78220cf06ae85f9d1db11dad9c3fd926799619eab3d28574aad  
db8cf328cd7aa
```

5.1.2 Verify Session

PG SOFT game system will call this operator API to check the player session token generated when players login to game.

Note:

- This API is required for HTML Scheme²⁴ authorization mode
- New player will be created automatically

Request

API URL : {OperatorAPIDomain}/VerifySession

HTTP method : POST

URL Parameters:

Parameter name	Data Type	Mandatory	Description
trace_id	String	Yes	Unique identity (GUID) of the request Note: Please set the parameter value as GUID format

Body Parameters:

Parameter name	Data Type	Mandatory	Description
operator_token	String	Yes	Unique identity of operator
secret_key	String	Yes	Shared passphrase between PG SOFT and operator
operator_player_session	String	Yes	Token generated by operator's system when the game is launched (HTML Scheme ²⁵)
game_id	Integer	No	Unique identity of games
ip	String	No	Player IP address
custom_parameter	String	No	<i>operator_param</i> values in HTML Scheme <i>extra_args</i> parameter.
bet_type	Integer	Yes	Game launch mode ²⁶

Example:

URL:

`https://operator.api/VerifySession?trace_id=b3f37e57-2873-40b1-aa95-f126c25ed311`

Body:

`operator_token=abc123&secret_key=a1b25cde5f3gh46ijkl&operator_player_session=a2b3c4d5e6f7g8&game_id=1&ip=1.1.1.1&custom_parameter=operatorparam123&bet_type=1`

²⁴ Please refer to [HTML Scheme](#) section for login flow and details

²⁵ Please refer to [HTML Scheme](#) section for more information

²⁶ Please refer to [Bet Types](#) section for complete list

Validation

Operator is required to perform validations to the following parameters:

Parameter name	Description	Expected behaviour
operator_token	Make sure the parameter value is the same as the one PG SOFT has provided.	Return error when not match
secret_key		
operator_player_session	Make sure the operator_player_session matches with the token generated in HTML Scheme ²⁷	

Response

Body Parameters:

JSON Object

Parameter name	Data type	Mandatory	Description
player_name	String	Yes	Unique identity of players - player name is not case sensitive - Only alphabet, number, and "@", "-", "_" symbols are allowed Note: Max 50 characters
nickname	String	No	Nickname of player Note: Max 50 characters
currency	String	Yes	Currency of player

Example:

```
{
  "data": {
    "player_name": "player1",
    "nickname": "nickname",
    "currency": "CNY"
  },
  "error": null
}
```

Error Response

Code	Message
1034	Invalid request
1200	Internal server error

²⁷ Please refer to [HTML Scheme](#) section for more information

5.1.3 Wallet Operation

5.1.3.1 Get Player Wallet

Operators are required to provide the total balance of all the wallets that can be used in PG SOFT games. This includes the total balance in the cash wallet.

Request

API URL : {OperatorAPIDomain}/Cash/Get

HTTP method : POST

URL Parameters:

Parameter name	Data Type	Mandatory	Description
trace_id	String	Yes	Unique identity (GUID) of the request Note: This ID is for issue troubleshooting only. Please do not use this ID for duplicate request validation.

Body Parameters:

Parameter name	Data Type	Mandatory	Description
operator_token	String	Yes	Unique identity of operator
secret_key	String	Yes	Shared passphrase between PG SOFT and operator
player_name	String	Yes	Unique identity of players • Player's name is not case sensitive
operator_player_session	String	No	Token generated by operator's system when the game is launched (HTML Scheme ²⁸)
game_id	Integer	No	Unique identity of games

Example:

URL:

https://operator.api/Cash/Get?trace_id=b3f37e57-2873-40b1-aa95-f126c25ed311

Body:

operator_token=abc123&secret_key=a1b25cde5f3gh46ijkl&player_name=player123&operator_player_session=a2b3c4d5e6f7g8&game_id=1

²⁸ Please refer to [HTML Scheme](#) section for more information

Validation

Operators are required to perform validation to the following parameters:

Parameter name	Description	Expected behaviour
operator_token	Make sure the parameter value is the same as the one PG SOFT has provided.	Return error when not match
secret_key		
player_name	Make sure the parameter value matches the given player's name in Verify Session ²⁹	

Response

Body Parameters:

JSON Object

Parameter name	Data type	Mandatory	Description
currency_code	String	Yes	Currency of player
balance_amount	Decimal	Yes	Cash Balance of player Note: • Amount must be ≥ 0 • Supports up to 2 decimal places • Extra decimal values will be truncated. Example: The game will show 11.12 if balance_amount value is 11.125 • Balance amounts do not included free game and bonus balances. • Operator should always return the latest player balance.
updated_time	Long	Yes	Updated time of player record (Unix time stamp in milliseconds)

Example:

```
{
  "data": {
    "currency_code": "CNY",
    "balance_amount": 11.12,
    "updated_time": 1530879795000
  },
  "error": null
}
```

²⁹ Please refer to [Verify Session](#) section for more information

Error Response

Code	Message
1034	Request is not valid
3005	Player wallet does not exist

5.1.3.2 Bet & Payout

PG SOFT system will expect **idempotent operations** for all API operations. Operators are required to perform duplicate request checks for all incoming transaction requests.

PG SOFT may resend a transaction due to network difficulties or other unforeseen circumstances. Please refer to the following precautions and process flow to correctly handle duplicated requests:

- Operators are required to validate both `operator_token` and `secret_key` for every transaction request.
- **transaction_id** is the only unique identifier for every transaction request, including bet and payout.
- In the case where a request with the same `transaction_id` is received and the previous request has already been processed successfully, please ignore the duplicated request, and return the previous successful response for the duplicated request.
- In the case where a request with the same `transaction_id` is received and the previous request has failed, please retry the operation and return the corresponding results to PG SOFT.
- For requests with a new `transaction_id`, operators may proceed to wallet addition or deduction accordingly and return the successful response to PG SOFT.

Real Transfer Amount

To prevent losses caused by incorrect currency base rate conversions, PG SOFT has introduced an update to the Seamless Wallet API. A new parameter `real_transfer_amount/realTransferAmount` has been added to both the API request and response.

The `real_transfer_amount` parameter represents the actual value of the amount in the request. Operators are required to include this parameter in the request and validate that the `realTransferAmount` returned in the API response matches the original value.

Operators should not update the players' wallet amount and must return an error response to PG SOFT if any discrepancies are found during validation.

Example:

If an operator uses a 1:1000 conversion for a currency with a base unit of 1000, and the `transfer_amount` in the request is 1, then the `real_transfer_amount` should be 1000.

If the operator receives an incorrect `real_transfer_amount`, they should return an error response and immediately inform PG SOFT.

Game Disconnected

If a player encounters an error and is disconnected from the game, the PG SOFT system will automatically resume the game from its previous stage (including free spin, bonus, or free game) without displaying any alert on the next player login.

Players will see the same screen as before the error occurred and will be able to continue the game from the previous stage.

Failed Transaction

May refer to [9.1.1 Fail the Request](#)

5.1.3.2.1 Bet Payout

This transaction request is the combination of both bet and payout in a single request. The PG SOFT system will call the operator's API in real time while the player is betting to either the add or deduct the player's balance. Operators are required to respond within 10 seconds or an API timeout will happen.

Request

API URL : {OperatorAPIDomain}/Cash/TransferInOut

HTTP method : POST

URL Parameters:

Parameter name	Data Type	Mandatory	Description
trace_id	String	Yes	Unique identity (GUID) of the request Note: Please set the parameter value as GUID format

Body Parameters:

Category	Parameter name	Data Type	Mandatory	Description
Authentication Information	operator_token	String	Yes	Unique identity of operator
	secret_key	String	Yes	Shared passphrase between PG SOFT and operator
	operator_player_session	String	No	Token generated by operator system when game is launched (HTML Scheme ³⁰)
General Bet Information	player_name	String	Yes	Unique identity of players • Player's name is not case sensitive
	game_id	Integer	Yes	Unique identity of games
	parent_bet_id	String	Yes	Unique identity of parent bet
	bet_id	String	Yes	Unique identity of child bet Note: Unique key for bets
	currency_code	String	Yes	Currency of player
	bet_amount	Decimal	Yes	Bet amount Note: PG SOFT will send a bet request with ZERO transfer amount for free spins
	win_amount	Decimal	Yes	Win amount. Note:

³⁰ Please refer to [HTML Scheme](#) section for more information

				PG SOFT will send a payout request with <u>ZERO</u> transfer amount for the following: • Every action (Example: split card) performed by players in card games • Special features in some slot games
	transfer_amount	Decimal	Yes	Player Win Loss amount Note: The amount can be positive or negative Negative: Balance deduction Positive: Balance addition
	real_transfer_amount	Decimal	Yes	[New] Actual value of the transaction amount in the real world Note: The operator is required to validate the value
	transaction_id	String	Yes	Unique identity of transaction. Format: {BetId}-{ParentBetId}-{transactionType}-{balanceId} Transaction type: 106: BetPayout 400: BonusToCash 403: FreeGameToCash Note: Max 50 characters
	wallet_type	String	No	To indicate the type of wallet for the current transaction. C: Cash B: Bonus G: FreeGame
	bet_type	Integer	Yes	Bet types of bet record: 1: Real game
	platform	String	No	Platform of player ³¹
	create_time	Long	Yes	Bet start time and bet last updated time

³¹ Please refer to [Platform](#) section for more information

	updated_time	Long	Yes	(Unix time stamp in milliseconds) Note: updated_time is not equal to bet end time
Bet Indicator	is_validate_bet	Boolean	No	To indicate if the request is a transaction that has been resent for validation True: Resent transaction False: Normal transaction
	is_adjustment	Boolean	No	To indicate if the request is an adjustment or a normal transaction for a reset game state True: Adjustment False: Normal transaction
	is_parent_zero_stake	Boolean	No	To indicate if the request is having 0 bet amount for the first bet of the round. True: Having 0 bet amount for the first bet of the round (For Bacarrat Deluxe's free hands action) False: Having amount larger than 0 in the first bet of the round
	is_feature	Boolean	No	To indicate the spin type True: Feature spin False: Normal spin
	is_feature_buy	Boolean	No	To indicate if a player has purchased the game feature Note: Only applicable for games with buy feature, such as feature buy, sure win bet etc
	is_end_round	Boolean	No	To indicate if the current game hand has ended
	free_game_transaction_id	String	No	Unique identity of free game. It can only be seen when Free Game has been assigned to the player Note: Only available for free game created using External API
Free Game Information	free_game_name	String	No	Given name of free game.

				It can only be seen when Free Game has been assigned to the player
	free_game_id	Integer	No	Unique identity of free game. It can only be seen when Free Game has been assigned to the player
	is_minus_count	Boolean	No	To indicate the spin type in free game True: Normal spin (Free game count is deducted) False: Free spin
Bonus Game Information	bonus_transaction_id	String	No	Unique identity of bonus game. It can only be seen when Bonus Game has been assigned to the player Note: Only available for bonus game created using External API
	bonus_name	String	No	Given name of bonus game. It can only be seen when Bonus Game has been assigned to the player
	bonus_id	Integer	No	Unique identity of bonus game. It can only be seen when Bonus Game has been assigned to the player
	bonus_balance_amount	Decimal	No	Total number of bonus amount in bonus wallet. It can only be seen when the player chooses to complete the bonus with cash.
	bonus_ratio_amount	Decimal	No	Rollover amount the player needs to achieve in bonus game. It can only be seen when the player chooses to complete the bonus with cash.

Example:

URL:

https://operator.api/Cash/TransferInOut?trace_id=b3f37e57-2873-40b1-aa95-f126c25ed311

Body:

```
operator_token=12345&secret_key=bfvcdb&player_name=player123&operator_player_session=vcbcvbcvb&game_id=1&parent_bet_id=123456&bet_id=123456&bet_type=1&currency_code=IDR&create_time=1530879795000&updated_time=1530879795000&bet_amount=100&win_amount=40&transfer_amount=-60&transaction_id=123456-123456-106-0&wallet_type=C&is_feature=False&is_minus_count=False&real_transfer_amount=-6000
```

Validation

Operator is required to perform validation to the following parameters:

Parameter name	Description	Expected behaviour
player_name	Make sure the parameter value matches the given player information in Verify Session ³²	Return error when not match
currency_code		
operator_token	Make sure the parameter value is the same as the one PG SOFT has provided.	
secret_key		Return successful response even with outdated operator_player_session
operator_player_session	Make sure the operator_player_session matches the token generated in HTML Scheme ³³ Do not validate operator_player_session when is_validate_bet or is_adjustment parameter value is true Note: PG SOFT system may resend request for validation or pending bet	
bet_amount	Make sure there is an error response when there is insufficient balance (bet_amount)	Return error response 3202
	Make sure there is an error response when there is insufficient balance (negative transfer_amount)	
transfer_amount	Make sure player balance after transaction (amount with 2 decimal places) is correct	Return successful response with the correct cash balance
	Make sure zero transfer amount is accepted for parent bet Note: PG SOFT system will send a request with <u>ZERO</u> transfer amount for Bacarrat Deluxe's free hands action	
	Make sure zero transfer amount is accepted for child bet Note: PG SOFT system will send a request with <u>ZERO</u> transfer amount for free spin	

³² Please refer to [Verify Session](#) section for more information

³³ Please refer to [HTML Scheme](#) section for more information

	Make sure there is an error response when $\text{win_amount} - \text{bet_amount} \neq \text{transfer_amount}$	Return error response 3073
real_transfer_amount	[New] Ensure an error response is triggered when an incorrect parameter value is detected. Please return an error if any of the following conditions are not met: - For 1:1 base unit currency: $\text{real_transfer_amount} = \text{transfer_amount}$ - For 1:1000 base unit currency and conversion handle by PG SOFT: $\text{real_transfer_amount} = \text{transfer_amount}$ - For 1:1000 base unit currency and conversion handle by operator (Default): $\text{real_transfer_amount} = \text{transfer_amount} * 1000$	Return error response 3107
transaction_id	Ignore wallet operation for duplicated requests (Requests with the same transaction_id)	Return successful response with the correct cash balance
parent_bet_id	Proceed wallet operation for request with the same game hand (Requests with the same parent_bet_id but different bet_id)	
bet_id		

Response

Body Parameters:

JSON Object

Parameter name	Data type	Mandatory	Description
currency_code	String	Yes	Currency of player
balance_amount	Decimal	Yes	Cash Balance of player Note: • Amount must be ≥ 0 • Support up to 2 decimal places • Extra decimal values will be truncated. Example: Game will show 11.12 if balance_amount value is 11.125
updated_time	Long	Yes	Updated time of transaction (Unix time stamp in milliseconds) Note: Response updated_time must be the same as the requested updated_time for transaction reference
real_transfer_amount	Decimal	Yes	[New] Actual value of the transaction amount in the real world Note: The operator is required to validate the value

Example:

```
{
  "data": {
    "currency_code": "IDR",
    "balance_amount": 11.12,
    "updated_time": 1530879795000
    "real_transfer_amount": -6000
  },
  "error": null
}
```

Error Response

Code	Message
3004	Player does not exist
3005	Player wallet does not exist
3021	Bet does not exist
3033	Bet failed
3107	Invalid configuration
3202	Insufficient player balance

Important Note:

Operators **SHOULD NOT** fail the requests for free spin or respin transactions. Please refer to [9.1.1 Fail the Request](#) for more information.

5.1.3.3 Balance Adjustment

PG SOFT will use this API to add or deduct a player's balance. This API may be used for certain events, such as Cash Tournaments.

Request

API URL : {OperatorAPIDomain}/Cash/Adjustment

HTTP method : POST

Body parameters:

Category	Parameter name	Data Type	Mandatory	Description
Authentication Information	operator_token	String	Yes	Unique identity of operator
	secret_key	String	Yes	Shared passphrase between PG SOFT and operator
General Information	player_name	String	Yes	Unique identity of players • Player's name is not case sensitive
	currency_code	String	Yes	Currency of player
	transfer_amount	Decimal	Yes	Adjustment Amount Negative amount: Balance deduction Positive amount: Balance addition
	real_transfer_amount	Decimal	Yes	[New] Actual value of the transaction amount in the real world Note: The operator is required to validate the value
	adjustment_id	String (100)	Yes	Reference ID for Adjustment
	adjustment_transaction_id	String (100)	Yes	Unique identity of transaction Format: {adjustment_id}-{transaction_type} Note: • Max 200 characters • Operator should use this parameter to check if

				the request is duplicated and implement idempotent operation
	adjustment_time	Long	Yes	Adjustment time (Unix time stamp in milliseconds)
	transaction_type	String	Yes	Adjustment source: 115: Money Rain Cash Reward (Promotion) 900: External Adjustment 901: Tournament Reward 902: Treasure Portal Cash Reward (Promotion)
	bet_type	Integer	Yes	Game launch mode ³⁴
	promotion_id	Integer	No	Reference ID for Promotion
	promotion_type	Integer	No	Promotion Type: 1: Money Rain 2: Money Rain Daily Bonus 3: Daily Cashpot 4: Treasure Portal
	remark	String	No	Remark for Balance Adjustment

Example :

operator_token=abc123&secret_key=a1b25cde5f3gh46ijkl&player_name=player123&bet_type=1¤cy_code=CN¥&transfer_amount=100.00&real_transfer_amount=100.00&adjustment_time=1530879795000&adjustment_id=ABC1530879&adjustment_transaction_id=ABC1530879-900&transaction_type=900

Validation

Operator is required to perform validation to the following parameters:

Parameter name	Description	Expected behaviour
player_name	Make sure the parameter value matches the given player information in Verify Session ³⁵	Return error when not match
currency_code		
operator_token		
secret_key	Make sure the parameter value is the same as the one PG SOFT has provided.	
transfer_amount	Make sure there is an error response when there is insufficient balance (negative transfer_amount)	Return error response 3202
	Make sure player balance after adjustment (amount with 2 decimal places) is correct	

³⁴ Please refer [Bet Types](#) for more info

³⁵ Please refer to [Verify Session](#) section for more information

	Make sure 0 or negative amount is accepted	Return successful response with correct balance
	Make sure the adjustment performs correctly and accordingly - Negative: Balance deduction - Positive: Balance addition	
adjustment_transaction_id	Ignore wallet operation for duplicated requests (Request with same transaction_id)	
real_transfer_amount	[New] Ensure an error response is triggered when an incorrect parameter value is detected. Please return an error if any of the following conditions are not met: - For 1:1 base unit currency: $\text{real_transfer_amount} = \text{transfer_amount}$ - For 1:1000 base unit currency and conversion handle by PG SOFT: $\text{real_transfer_amount} = \text{transfer_amount}$ - For 1:1000 base unit currency and conversion handle by operator (Default): $\text{real_transfer_amount} = \text{transfer_amount} * 1000$	Return error response 3107

Response

Body Parameters:

JSON Object

Parameter name	Data type	Mandatory	Description
adjust_amount	Decimal	Yes	Adjusted Amount Note: - This amount is for reference purpose. PG SOFT will not validate it. - Response adjust_amount must be the same as the requested transfer_amount for transaction reference
balance_before	Decimal	Yes	Player balance before adjustment (balance_before) & Player balance after adjustment (balance_after) Note: - Amount must be ≥ 0 - Only support up to 2 decimals
balance_after	Decimal	Yes	

			• Extra decimal values will be truncated. Example: Game will show 11.12 if balance_amount value is 11.125
updated_time	Long	Yes	Updated time of transaction (Unix time stamp in milliseconds) Note: Response updated_time must be the same as the requested adjustment_time for transaction reference
real_transfer_amount	Decimal	Yes	[New] Actual value of the transaction amount in the real world Note: The operator is required to validate the value

Example:

```
{
  "data": {
    "adjust_amount": 100.00,
    "balance_before": 0.00,
    "balance_after": 100.00,
    "updated_time": 1737941479768
    "real_transfer_amount": 100.00
  },
  "error": null
}
```

Error Response

Code	Message
3004	Player does not exist
3005	Player wallet does not exist

5.1.3.4 Update Bet Details

PG SOFT game system will call this operator API to update the operator's latest bet end time when the bet is successful in PG SOFT system.

Request

API URL : {OperatorAPIDomain}/Cash/UpdateBetDetail

HTTP method : POST

Body Parameters:

Parameter name	Data Type	Mandatory	Description
operator_token	String	Yes	Unique identity of operator
secret_key	String	Yes	Shared passphrase between PG SOFT and operator
bet_details	Json Array	Yes	List of bet ID with bet end time
updated_time	Long	Yes	API request time (Unix time stamp in milliseconds)

Parameter: bet_details

Parameter name	Data Type	Mandatory	Description
bet_id	String	Yes	Unique identity of child bet Note: Unique key for bets
end_time	Long	Yes	Bet end time (Unix time stamp in milliseconds)

Example:

```
operator_token=abc123&secret_key=a1b25cde5f3gh46ijkl&bet_details=[{"bet_id":"1456887894","end_time":"1644476958000"}, {"bet_id":"1456887895","end_time":"1644476958000"}]&updated_time=1645414794000
```

Validation

Operator is required to perform validation to the following parameters:

Parameter name	Description	Expected behaviour
operator_token	Make sure the parameter value is the same as the one PG SOFT has provided.	Return error when not match
secret_key		

Response

Body Parameters:

JSON Object

Parameter name	Data Type	Mandatory	Description
is_success	Boolean	Yes	To indicate if the request is received True: Received False: Not Received

Example:

```
{
  "data": {
    "is_success": true
  },
  "error": null
}
```


5.2 PG SOFT Provided API

5.2.1 API Formats

5.2.1.1 Request

The API uses HTTP form methods and a RESTful endpoint structure. Operator is required to send requests in the following content type:

Content-Type: `application/x-www-form-urlencoded`

5.2.1.2 Response

API returns JSON-formatted responses. For successful and failed API requests, PG SOFT system will return an API response with HTTP status code 200.

API uses global JSON response format for all API methods, with the following content type:

Content-Type: `application/json`

Parameters:

Parameter name	Data Type	Description
<code>data³⁶</code>	JSON Object	Response information for API methods
<code>error³⁷</code>	JSON Object	Error information when an error or exception occur

Example:

Successful Response

```
{
  "data": {
    [API method response. JSON object format may vary depending on API methods]
  },
  "error": null,
}
```

Failed/Error Response

```
{
  "data": null,
  "error": {
    "code": "[Error code]",
    "message": "[Error message]"
  }
}
```

³⁶ This field will show *null* value in error response

³⁷ This field will show *null* value in successful API response. Please refer to [Error Handling](#)

5.2.2 Bet History Interpreter (Optional)

To reveal a player's bet results and details for specific bet.

Step 1

Before getting the bet detail, operator is required to call this API to get operator's session (Session timeout: 30 minutes).

Request

API URL : {PgSoftAPIDomain}/Login/v1/LoginProxy

HTTP method : POST

URL Parameters:

Parameter name	Data Type	Mandatory	Description
trace_id	String	Yes	Unique identity (GUID) of the request Note: Please set the parameter value as GUID format

Body Parameters:

Parameter name	Data Type	Mandatory	Description
operator_token	String	Yes	Unique identity of operator
secret_key	String	Yes	Shared passphrase between PG SOFT and operator

Example:

URL:

https://api.pg-bo.me/external/Login/v1/LoginProxy?trace_id=b3f37e57-2873-40b1-aa95-f126c25ed311

Body:

operator_token=abc123&secret_key=a1b25cde5f3gh46ijk1

Response

Body Parameters:

JSON Object

Parameter name	Data Type	Description
operator_session	String	Operator session

Example:

```
{
  "data": {
    "operator_session": E4ECB5CF-0BC6-4DA8-8551-8526F48AB9CD
  },
  "error": null
}
```

Step 2

Get bet detail by:

API URL : {PgSoftAPIDomain}/external-game-launcher/api/v1/GetLaunchURLHTML

HTTP method : POST

URL Parameters:

Parameter name	Data Type	Mandatory	Description
trace_id	String	Yes	Unique identity (GUID) of the request Note: Please set the parameter value as GUID format

Body Parameters:

Parameter name	Data Type	Mandatory	Description
path	varchar (500)	Yes	Bet History Interpreter path: /history/redirect.html
extra_args	Query String Paramater	Yes	Bet history interpreter parameter (Please refer to relevant parameter list) Note: - Please encode the value with UrlEncode to avoid unexpected errors Maximum 500 characters
url_type	varchar (50)	Yes	URL Type: web-history: Bet History interpreter
client_ip	varchar (50)	Yes	Player IP
group_id	UInt16	No	Operator group ID

extra_args Parameters:

Parameter name	Data Type	Mandatory	Description
t	String	Yes	Operator session
psid	String	Yes	Parent Bet ID
sid	String	Yes	Bet ID
gid	String	No	Game ID
lang	String	No	Language: - en (default) - zh
type	String	Yes	Fixed value: operator

Example:

Body:

```
path=/history/redirect.html&extra_args=t%3DE4ECB5CF-0BC6-4DA8-8551-8526F48AB9CD%26psid%3D1821471534863220736%26sid%3D1821471534863220736%26lang%3Den%26type%3Doperator&url_type=web-history&client_ip=1.1.1.1
```

Response

HTTP Body response:

The browser only needs to execute the HTML page to launch the game.

*** Note: The HTML code is subject to adjustment and optimization and does not follow a fixed format. Please avoid using deserialization or parsing methods to retrieve the final game launch URL.**

HTTP status code response:

Status Code	Description	Reason	Solution
200	Successful response	-	-
400	Invalid Parameters	The format of the parameter is not provided as requested	Check whether the HTTP Method, HTTP Header, and HTTP Body are requested accordingly to the requirements.
404	Path not found	Invalid path	Check whether the path is correct.
500	Server Internal Error	An internal server error occurred	• Retry • Inform PG SOFT
504	Gateway Timeout	The request has failed to reach the destination server successfully and a proxy server failed.	• Retry • Inform PG SOFT

Example for HTML Response (For reference only):

```
<!doctype html><html lang="en">...</html>
```

5.2.3 Others (Optional)

PG SOFT provides several APIs for different modules that are optional for integration. Operator may choose to integrate based on their requirements:

- **Bet History** : To get a player's bet history.
- **Game List** : To get the latest game list and available bet sizes.
- **Free Game** : To perform several actions (For example: Create free game, assign player into free game, cancel free game, etc.) through API.
- **Bonus Game** : To perform several actions (For example: Create bonus game, assign player into bonus game, cancel bonus game, etc.) through API.
- **Tournament** : To get tournament list, add player into tournament and get tournament ranking through API.

5.3 Integrate PG SOFT Game into Operator's iOS Application

Operator's application needs to meet the following requirements to launch PG SOFT games in the application client:

- iOS 8.0 or above
- WKWebView

PG SOFT game will prompt a "Poor Visit Experience" message if it does not meet the minimum requirements. Players are allowed to continue launching the game but it may not guarantee a stable, and high performance experience of the PG SOFT game.

To ensure the application client passes the PG SOFT game requirement check, operator is required to include the following codes when initializing WKWebView component in application:

```
WKUserContentController *controller = [[WKUserContentController alloc] init];  
[controller addScriptMessageHandler: self name: @"Could be any srting value"];  
  
WKWebViewConfiguration *configuration = [[WKWebViewConfiguration alloc] init];  
configuration.userContentController = controller;  
  
WKWebView *webView = [[WKWebView alloc] initWithFrame: CGRectZero configuration:  
configuration];  
webView.uiDelegate = self;  
self.view = webView;
```

This is to construct a WKUserContentController object and set its userContentController property on the configuration. Please take note that the userContentController must have at least one scriptMessageHandler (assign through addScriptMessageHandler)

For more details regarding WKWebView, please refer to [Apple Developer Documentation](#)

6 Resolve Pending Bet

Sometimes pending bets may occur due to unforeseen circumstances during gameplay due to network latency, system failure, system timeout and so on.

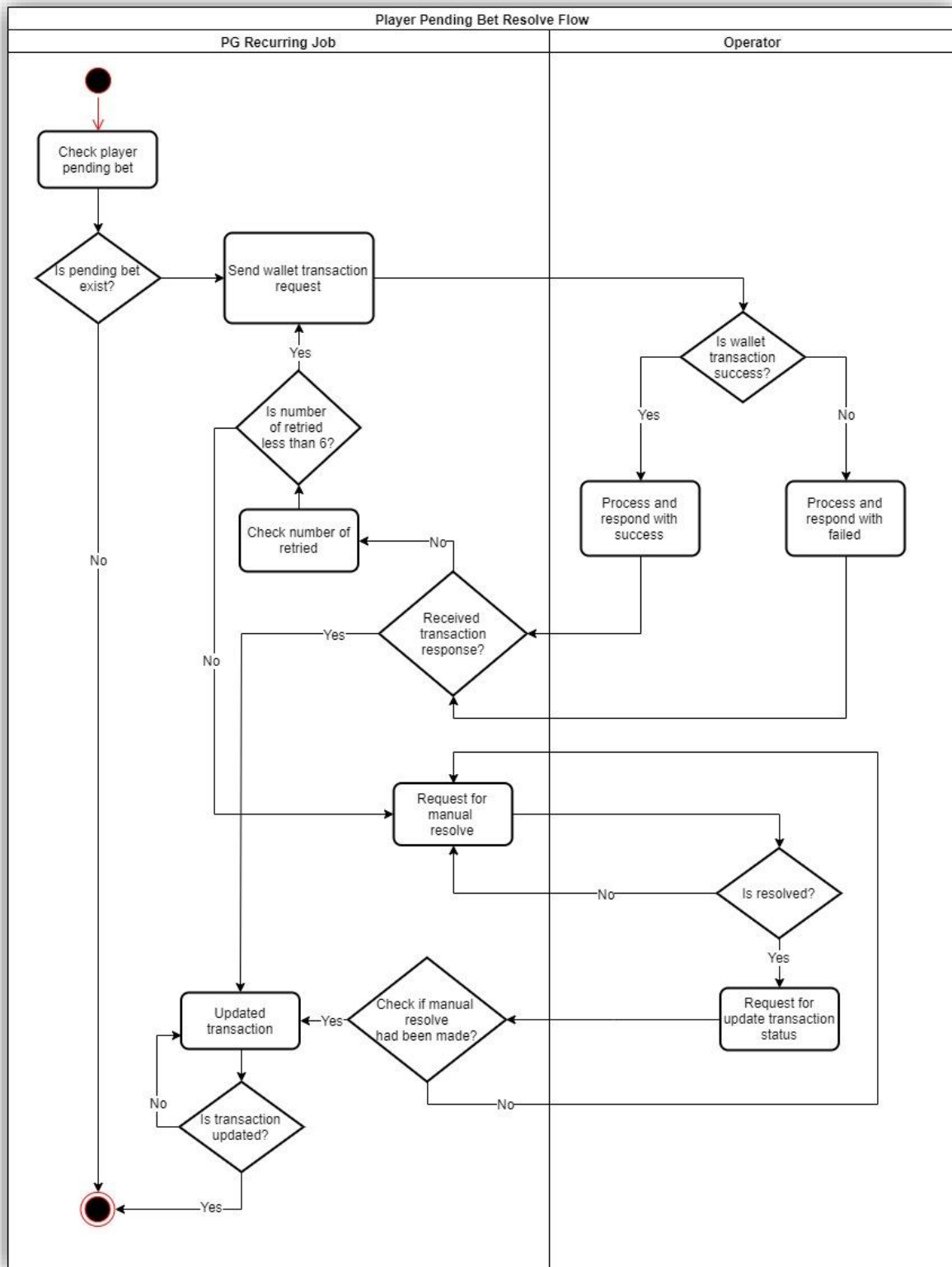
Generally, pending bet will be resolved automatically when the player relaunches the game. However, the bet will remain in pending status for some rare scenarios:

- Player did not relaunch the game after encountering an error.
- PG SOFT system did not receive any response from the operator when resolving pending bets.
- PG SOFT system received an error response from the operator when resolving pending bets.

Default interval of pending bet resolving time is set to 60 minutes. Pending bets which exceed the interval will be resolved by PG SOFT system automatically. This interval is customizable, and every operator can have their own bet resolve interval time.

Manual resolve is required when pending bets cannot be resolved after several attempts. In such cases, the operator is required to resolve the pending bet manually, and PG SOFT will mark these bets statuses as complete after getting the confirmation from the operator.

Screenshot



Player Pending Bet Resolve Flow

7 Reset Game State

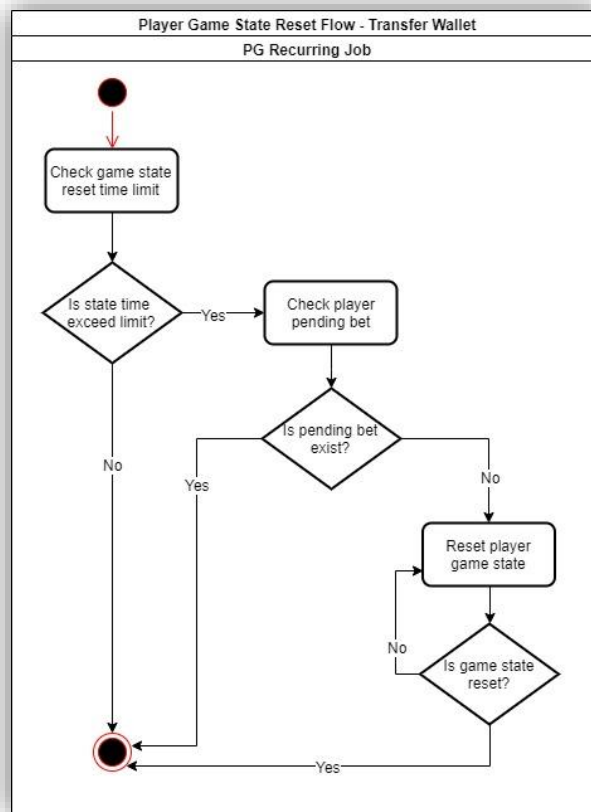
All games will support the resume feature which allows returning players to continue the game at their last game state from any device. For example, players will be able to continue a game with a bonus game state few days after even if the player had closed the game when the player won the bonus game.

However, the player's game state will reset every 90 days by default. This interval is customizable, and every operator can have their own game state reset interval time. System will clear all finished and unfinished game progress, and the player will not be able to continue a game after it has been reset.

For transfer wallet mode, every player game state will reset every 90 days by default:

- PG SOFT system will reset a player's game state automatically if there are no pending bets for the player.

Screenshot



Player's Game State Reset Flow for Transfer Wallet

8 Restrictions

8.1 IP

Players from the following restriction area are not able to access PG SOFT game:

- Malaysia
- Singapore
- Taiwan
- United States of America
- Macau
- Israel
- Iran
- North Korea
- Australia
- United Kingdom

8.2 Currency

Please find the following for the currencies not supported by PG SOFT:

- AUD
- ESP
- HRK
- STD
- TRL
- HKD
- GBP
- JPY (Not accepted for new requests)
- MOP
- MYR
- SGD
- TWD
- All cryptocurrencies except for crypto currencies in the Currencies' Appendix

9 Appendix

9.1 Error Handling

9.1.1 Fail the Request

Operators should always practice idempotent responses to prevent duplicate modifications to player balances. If one of the errors below has already been returned in a previous response, operators are required to return the same error response again to ensure the request status remains synchronized between the operator and PG SOFT.

The following error codes will cause PG SOFT games to stop and display an error during bet transactions. The bet request will be cancelled when any of the following error codes are returned:

- 3004 PlayerNotFoundException
- 3005 WalletNotFoundException
- 3008 BonusNotFoundException
- 3009 FreeGameNotFoundException
- 3019 NotEnoughFreeGameException
- 3021 BetNotFoundException
- 3030 FreeGameAlreadyExpiredException
- 3035 InvalidFreeGameMultiplierException
- 3042 BonusLockedException
- 3043 FreeGameLockedException
- 3055 SnapshotNotFoundException
- 3057 InvalidFreeGameCoinSizeException
- 3059 BetLimitExceededException
- 3062 TransactionRolledBackException
- 3065 BonusAlreadyExpiredException
- 3073 BetFailedException
- 3094 MaximumBetLimitReachErrorCode
- 3095 BalanceDecreaseLimitReachErrorCode
- 3103 PromotionTransactionNotFoundException
- 3200 NotEnoughBalanceException
- 3201 NotEnoughBonusBalanceException
- 3202 NotEnoughCashBalanceException
- 3250 – 3259 reserved error codes

However, the above error codes should only be used for bet or transfer-out requests. Operators SHOULD NOT return these error codes under the following conditions, as doing so may cause incorrect game behaviour and results, leading to operator losses or significant player winnings:

- Payout request
- Feature spin / free spins / Respins (when parameter `is_feature = true`)

9.2 Error codes

Error code	Description
1034	Request is not valid
1035	Operation Failed
1200	Internal server error
1204	Failed to get operator
1300	Invalid player session
1301	Player session token is empty
1302	Invalid player session
1303	Server error occurs
1305	Invalid player – Player does not exist: [player name]
1306	Player is blocked to access current game
1307	Invalid player session
1308	Player session is expired
1309	Player is inactive
1310	Failed to verify operator player session
1315	Player's operation in progress
1400	Game is under maintenance
1401	Game is inactive
1402	Game does not exist
3001	Value cannot be null
3004	Player does not exist
3005	Player wallet does not exist
3006	Player wallet already exists
3009	Free game does not exist
3013	Service Exception: Not enough balance to transfer
3014	Free game cannot be cancelled
3019	Not enough free game
3021	No bet exists
3022	Bet already pay-out
3030	Free game expired
3031	Free game already converted
3032	Bet already existed
3033	Bet failed
3034	Pay-out failed
3035	Invalid multiplier
3036	Not enough balance to convert
3040	Transaction does not exist
3202	Not enough cash balance to bet

9.3 Currencies

Currency Code	Currency Symbol	Currency Name	Base Unit
AED	د.إ.	Dirham des Émirats arabes unis	1
AFN	ؑ	Afghanistan Afghani	1
ALL	L	Albanian Lek	1
AMD	֏	Armenian Dram	1
ANG	ƒ	Netherlands Antilles Guilder	1
AOA	Kz	Angola Kwanza	1
ARS	\$	Argentine Peso	1
ARSB	\$	ARS blue dollar	1
AWG	ƒ	Aruba Guilder	1
AZN	₼	Azerbaijani Manat	1
BAM	KM	Bosnian Convertible Marka	1
BBD	\$	Barbados Dollar	1
BDT	Tk	Taka bangladais	1
BGN	лв	Lev bulgare	1
BHD	BD	Bahrain Dinar	1
BIF	FBu	Burundian Franc	1000
BMD	\$	Bermuda Dollar	1
BND	B\$	Brunei Dollar	1
BOB	\$b	Bolíviano bolivien	1
BRL	R\$	Brazil Real	1
BSD	\$	Bahamian Dollar	1
BTN	Nu.	Bhutanese Ngultrum	1
BWP	P	Botswana Pula	1
BYN	BYN	Belarusian Rouble	1
BYR	Br	Belarusian Ruble	1
BZD	BZ\$	Belize Dollar	1
CAD	C\$	Canadian Dollar	1
CDF	FC	Congolese Franc	1000
CHF	Fr.	Swiss Franc	1
CLP	\$	Chilean Peso	1
CNY	¥	Chinese Yuan	1
COP	\$	Colombian Peso	1000
CRC	₡	Colon costaricain	1
CSD	din	Serbian Dinar	1
CUP	₱	Cuba Peso	1

CVE	\$	Cape Verde Escudo	1
CZK	Kč	Czech Koruna	1
DJF	Fdj	Djiboutian Franc	1
DKK	kr.	Danish Krone	1
DOP	RD\$	Peso dominicain	1
DZD	DA	Dinar algérien	1
EGP	£	Livre égyptienne	1
ERN	Nkf	Eritrea Nakfa	1
ETB	ብር	Ethiopian Birr	1
EUR	€	Euro	1
FJD	\$	Fiji Dollar	1
FKP	£	Falkland Islands (Malvinas) Pound	1
GEL	ლ	Georgian Lari	1
GHS	GH¢	Ghanaian Cedi	1
GIP	£	Gibraltar Pound	1
GMD	D	Gambian Dalasi	1
GNF	FG	Guinean Franc	1000
GTQ	Q	Quetzal guatémaltèque	1
GYD	\$	Guyana Dollar	1
HNL	L	Honduran Lempira	1
HTG	G	Haiti Gourde	1
HUF	Ft	Hungarian Forint	1
IDR	Rp	Indonesian Rupiah	1000
ILS	₪	Shekel israélien	1
INR	₹	Indian Rupee	1
IQD	د.ع	Dinar irakien	1000
IRR	₪	Iranian Rial	1000
ISK	kr	Couronne islandaise	1
JMD	J\$	Jamaica Dollar	1
JOD	د.أ	Jordanian Dinar	1
KES	KSh	Kenyan Shilling	1
KGS	KGS	Som du Kirghizistan	1
KHR	៛	Cambodian Riel	1000
KMF	CF	Comorian Franc	1
KPW	₩	North Korea Won	1
KRW	₩	South Korean Won	1000
KWD	ك.د	Kuwait Dinar	1
KYD	\$	Cayman Islands Dollar	1
KZT	₸	Kazakhstani Tenge	1
LAK	₭	Lao Kip	1000

LBP	ل.ل	Livre libanaise	1000
LKR	Rs	Roupie sri lankaise	1
LRD	\$	Liberia Dollar	1
LSL	L	Lesotho Loti	1
LVL	Ls	Latvian Lats	1
LYD	LD	Dinar libyen	1
MAD	DH	Moroccan Dirham	1
MBTC	mB	Milli Bitcoin	1
MDL	L	Moldovan Leu	1
MGA	Ar	Madagascar Ariary	1000
MKD	ден	Dinar macédonien	1
MMK	K	Burmese Kyat	1000
MNT	₮	Mongolian Tughrik	1000
MUR	Rs	Mauritius Rupee	1
MVR	Rf.	Maldivian Rufiyaa	1
MWK	MK	Malawian Kwacha	1
MXN	\$	Mexican Peso	1
MZN	MT	Mozambican Metical	1
NAD	\$	Namibia Dollar	1
NGN	₦	Nigerian Naira	1
NIO	C\$	Cordoba nicaraguayen	1
NOK	kr	Norwegian Krone	1
NPR	Rs	Nepal Rupee	1
NZD	\$	New Zealand dollar	1
OMR	ر.ع.	Oman Rial	1
PAB	B/.	Balboa panaméen	1
PEN	S/	Peruvian Sol	1
PGK	K	Papua New Guinea Kina	1
PHP	₱	Philippine Peso	1
PKR	Rs	Pakistan Rupee	1
PLN	zł	Poland Złoty	1
PYG	Gs	Guarani paraguayen	1000
QAR	QR	Rial qatari	1
RON	lei	Romanian Leu	1
RSD	РСД	Serbian Dinar	1
RUB	₽	Russian Ruble	1
RWF	FRw	Rwandan Franc	1000
SAR	SR	Rial saoudien	1
SBD	\$	Solomon Islands Dollar	1
SCR	SR	Seychellois Rupee	1
SDG	ج.س.	Livre soudanaise	1

SEK	kr	Swedish Krona	1
SHP	£	Saint Helena Pound	1
SLE	Le	Sierra Leonean Leone	1
SLL	Le	Sierra Leonean Leone	1000
SOS	S	Somalia Shilling	1
SRD	\$	Suriname Dollar	1
SVC	\$	Colon salvadorien	1
SYP	£	Livre syrienne	1
SZL	L	Swazi Lilangeni	1
THB	฿	Thai Baht	1
TJS	TJS	Tajikistan Somoni	1
TMT	T	Turkmenistan Manat	1
TND	د.ت	Dinar tunisien	1
TOP	T\$	Tongan pa'anga	1
TRX	TRX	TRON Crypto	1
TRY	₺	Turkish Lira	1
TTD	TT\$	Trinidad and Tobago Dollar	1
TUSD	TUSD	TrueUSD(Crypto)	1
TZS	TSh	Tanzanian Shilling	1000
UAH	₴	Ukrainian Hryvnia	1
UBTC	μB	Micro Bitcoin	1
UGX	USh	Uganda Shilling	1000
USD	\$	United States Dollar	1
USDC	USDC	USD Coin (Crypto)	1
USDT	₮	Tether	1
UYU	\$U	Peso uruguayen	1
UZS	so'm	Uzbekistani Som	1000
VES	Bs.S	Venezuelan Bolívar	1
VND	₫	Vietnamese Dong	1000
VUV	VT	Vanuatu Vatu	1
WST	\$	Samoa Tala	1
XAF	FCFA	Central African CFA Franc BEAC	1
XCD	\$	East Caribbean Dollar	1
XOF	CFA	CFA Franc	1
XPF	₣	CFP Franc	1
YER	ﷲ	Rial yéménite	1
ZAR	R	Rand sud-africain	1
ZMW	ZK	Zambian Kwacha	1
GC	GC	Stake Gold Coins (Social Casino Currency)	1
SC	SC	Stake Cash (Social Casino Currency)	1

9.4 Currency Base Unit & Handling

Operators should always verify the base unit in the currency list before submitting a request for environment setup. Below is a list of currencies with a base unit of 1000. Operators are required to choose whether the conversion will be handled by PG SOFT or the operator in the integration application.

- BIF (Burundian Franc)
- CDF (Congolese Franc)
- COP (Colombian Peso)
- GNF (Guinean Franc)
- IDR (Indonesian Rupiah)
- IQD (Dinar Iranian)
- IRR (Iranian Rial)
- KHR (Cambodian Riel)
- KRW (South Korean Won)
- LAK (Lao Kip)
- LBP (Livre Lebanese)
- MGA (Madagascar Ariary)
- MMK (Burmese Kyat)
- MNT (Mongolian Tughrik)
- PYG (Guarani Paraguay)
- RWF (Rwandan Franc)
- SLL (Sierra Leonean Leone)
- TZS (Tanzanian Shilling)
- UGX (Uganda Shilling)
- UZS (Uzbekistani Som)
- VND (Vietnamese Dong)

Note:

Please refer to the appendix for the complete currency list with base units. Always check with PG SOFT if a currency is not listed.

The table below explains the relationship between currencies with a base unit of 1000 and how this impacts the API, system, reporting, and settlement processes:

API		Get	TransferIn TransferOut TransferInOut	Adjustment	External- Grabber
Real Amount		1000			
Game Display/ PG SOFT BackOffice		1			
Settlement		1000			
Operator Handle 1:1000 Conversion (Default)	PG SOFT Send		1	1	1
	Operator Send	1			
PG Handle 1:1000 Conversion	PG Send		1000	1000	1
	Operator Send	1000			

9.5 Bet Types

Bet Type	Game Mode
1	Real game

9.6 Transaction Types

Code	Transaction type
1	Cash
2	Bonus game
3	Free game

9.7 Platforms

Code	Category	Platform
1	Web	Windows
2		macOS
3		Android
4		iOS
5		others
6	Cordova	Android
7		iOS
8	Electron	Windows
9		macOS
10	Native	Windows
11		macOS
12		Android
13		iOS
98	others	System
99		Smartbot

9.8 Languages

Code	Language
en	English (default)
en-social	Social Language English
en-stkus	Social casino (for currencies SC & GC)
ar-EG	Arabic
az-Latn-AZ	Azerbaijani (Latin)
bg-BG	Bulgarian
bn-BD	Bengali
cs-CZ	Czech
da-DK	Danish
de-DE	German
el-GR	Greek
es-ES	Spanish
et-EE	Estonian
fa-IR	Farsi
fi-FI	Finnish
fr-FR	French
hi-IN	Hindi
hu-HU	Hungarian
hy-AM	Armenian
id-ID	Indonesian
it-IT	Italian
ja-JP	Japanese
ko-KR	Korean
lo-LA	Lao
lt-LT	Lithuanian
mn-MN	Mongolian
my-MM	Burmese
no-NO	Norwegian
nl-NL	Dutch
pl-PL	Polish
pt-PT	Portuguese
pt-BR	Portuguese Brazilian
ro-RO	Romanian
ru-RU	Russian
sr-Latn-RS	Serbian (Latin)
si-LK	Sinhala
sk-SK	Slovak

sq-AL	Albanian
sv-SE	Swedish
th-TH	Thai
tr-TR	Turkish
uk-UA	Ukrainian
ur-PK	Urdu
uz-Latn-UZ	Uzbek (Latin)
vi-VN	Vietnamese
zh	Chinese